

Inhibitory circuits control leg movements during *Drosophila* grooming

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eLife Assessment

Using a combination of connectomics, optogenetics, behavioral analysis and modeling, this study provides **important** findings on the role of inhibitory neurons in the generation of leg grooming movements in *Drosophila*. The data as presented provide **convincing** evidence that the identified neuronal populations are key in the generation of rhythmic leg movements. Based on reconstructions from ventral nerve cord electron microscopy data, the authors uncover distinct pathways to the motor neurons, which they propose inhibit and disinhibit antagonistic sets of motor neurons. This results in an alternation of flexion and extension. By analyzing limb kinematics upon silencing of specific populations of premotor inhibitory neurons and using computational modelling, they show the potential role of these neurons in rhythmic leg movement. The work will interest neuroscientists and particularly those working on motor control.

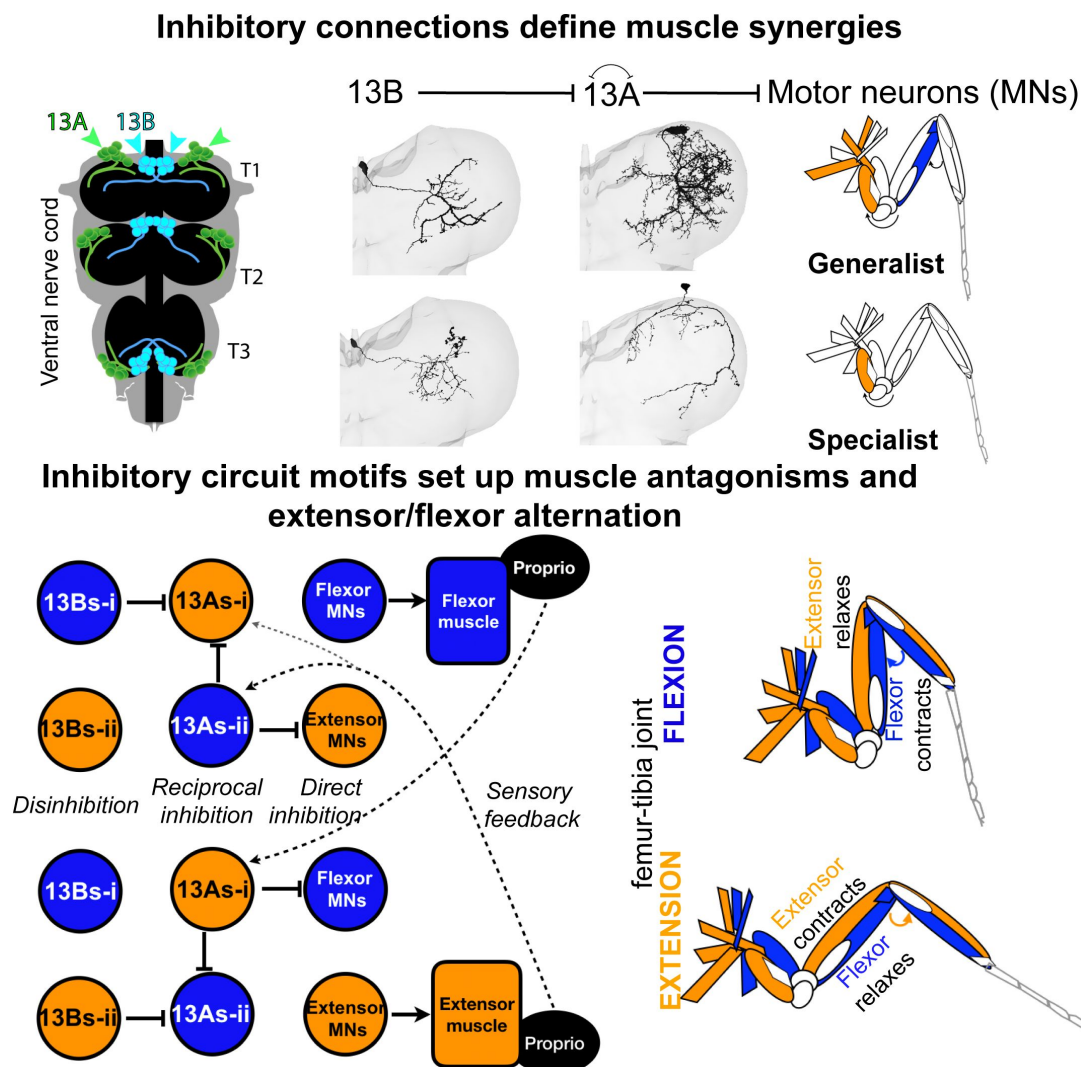
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Abstract

Limbs execute diverse actions coordinated by the nervous system through multiple motor programs. The basic architecture of motor neurons that activate muscles which articulate joints for antagonistic flexion and extension movements is conserved from flies to vertebrates. While excitatory premotor circuits are expected to establish sets of leg motor neurons that work together, our study uncovered an instructive role for inhibitory circuits — including their ability to generate rhythmic leg movements. Using electron microscopy data in the *Drosophila* nerve cord, we categorized ~120 GABAergic inhibitory neurons from the 13A and 13B hemilineages into classes based on similarities in morphology and connectivity. By mapping their connections, we uncovered pathways for inhibiting specific groups of motor neurons, disinhibiting antagonistic counterparts, and inducing alternation between flexion and extension. We tested the function of specific inhibitory neurons through optogenetic activation and silencing, using high resolution quantitative analysis of leg movements during grooming. We combined findings from anatomical and behavioral analyses to construct a computational model that can reproduce major aspects of the

observed behavior, demonstrating that these premotor inhibitory circuits can generate rhythmic leg movements.

Graphical abstract



Introduction

All animals with limbs face the challenge of coordinating their movements to achieve precise motor control. Despite a limited set of muscles in each limb, the nervous system produces multiple flexible actions to generate behavior. These movements rely on a balance of inhibition and excitation, though the specific circuitry remains unclear. In various insects, coordinated leg movements have been studied extensively during walking and grooming, revealing both the flexibility and stereotypy of action sequences^{1,2,3,4,5}. These studies highlight how insects provide a powerful system to understand the circuit basis of limb coordination. *Drosophila* grooming involves coordinated, rhythmic leg movements to sweep the body and remove debris⁶ with different actions prioritized sequentially. While sensory, command-like, and motor neurons (MNs) are known components of grooming circuits^{7,8,9,10,11,12,13}, the contribution of GABAergic inhibitory neurons has not been systematically investigated. We hypothesize that these neurons contribute to limb coordination and subroutine selection.

Leg movements during grooming require precise flexor-extensor coordination, controlled by MNs and premotor circuits. Flies use 14 intrinsic leg muscles and 3-5 body wall muscles^{14,15,16}, organized into antagonistic pairs. Around 70 excitatory MNs innervate these muscles^{16,17,18,19,20} and recent connectomic mapping has revealed premotor neurons in the ventral cord neuromeres associated with each leg^{21,22}. There are around 622 local premotor interneurons²¹, suggesting complex control architectures. Studies in stick insects, locusts, cockroaches, and crustaceans, have also revealed large numbers of interconnected premotor interneurons, some of which constitute central pattern generator circuits that shape rhythmic movements and interlimb coordination^{5,23,24,25,26,27,28,29,30,31,32,33}. In *Drosophila* larvae, circuits containing both excitatory and inhibitory interneurons generate rhythmic muscle contractions within segments and propagate peristaltic waves between segments for crawling^{34,35,36,37,38,39}. Together, these findings support the idea that premotor neurons may be a core component for generating and coordinating rhythmic motor control.

We investigate the role of inhibitory neurons in coordinating which leg muscles in adult *Drosophila* work together or antagonistically, and how they might produce rhythmic alternations. Neurons from a given lineage usually share a neurotransmitter, and there are 12 GABAergic hemilineages present in the ventral nerve cord (VNC)⁴⁰. We identified neurons from GABAergic 13A and 13B hemilineages in a behavioral screen for grooming defects. Approximately 67 ± 6 13A neurons and 47 ± 1 13B neurons have been reported per hemisegment^{40,41}. While activating all 13B and some 13A neurons induces leg extensions^{40,42}, further investigation is required to understand specific role of different subsets in leg coordination. The role of 13A neurons has been unclear due to the lack of tools for exclusive genetic labeling.

Since grooming and walking⁴³ require leg movements, we expect neural control circuits to overlap, particularly in flexion-extension alternation. MNs controlling similar muscles within a joint receive similar premotor inputs²¹. Descending neurons hypothesized to be involved in walking synapse onto premotor inhibitory neurons from several lineages²². These central circuits also receive sensory feedback from leg proprioceptors^{44,45,46,47}. Recent connectome mapping and genetic tools provide opportunities to test how central and peripheral signals coordinate limb movement.

Muscle synergies describe groups of co-activated muscles, while motor primitives are elemental movement patterns that serve as building blocks of behavior^{48,49}. Micro-stimulation of specific spinal cord regions in vertebrate models induced coordinated muscle contractions^{50,51,52,53,54,55,56}, with co-stimulation leading to combinations of contractions, indicating how coordinated regulation by premotor circuits can simplify assembly of more complex movements^{49,56,57,58}. Additional evidence for synergies comes from electromyography, kinematics, neural recordings and computational modeling^{59,60,61,62,63,64,65,66,67}. In vertebrates, the neural circuits responsible for the coordinated limb movements include motor pools, a topographic map, and commissural neurons^{68,69,70,71,72,73,74}. The activation of muscles controlling multiple joints (synergies) have been primarily described in terms of excitation^{49,55}. Muscle synergies may simplify motor control in insect walking and flight^{2,75,76,77,78,79}.

We hypothesize that similar synergies could simplify control of the rhythmic leg movements during grooming in flies and that inhibitory circuits play a critical role. We find that inhibitory neurons target different groups of MNs, providing an alternative way to construct muscle synergies. We demonstrate that normal activity of these inhibitory neurons is essential for the rhythmic coordination of leg flexion and extension during grooming. Knocking down inhibitory receptors on MNs in flies reduces locomotion speed⁸⁰, but the specific inhibitory neurons involved remain unidentified. By analyzing limb kinematics in grooming flies and silencing

specific 13A and 13B neurons, we demonstrate their critical role in spatial and temporal limb coordination. Our findings suggest inhibitory circuits play a broader role in coordinated and rhythmic limb movements.

Results

We describe how inhibitory 13A and 13B neurons affect grooming. We categorize them by morphology, and map their connectivity patterns. We present evidence for muscle synergies and a role for 13A/B neurons in coordinating rhythmic limb movements. Finally, we generate a computational model to integrate and simulate our findings.

Inhibitory Interneurons in 13A and 13B Hemilineages Affect Grooming Behavior

Broad optogenetic activation of inhibitory neurons causes freezing, while activation of fewer neurons in the hemilineages 13A and 13B results in reduced grooming behavior and poor leg coordination, including static over-extension of front legs in clean or dusted flies (**Figures 1D**, **1F** and **1F'**, **Figure 1—Video 1**). We conclude that the critical neurons are located in the ventral cord because most Split GAL4 lines only express in the VNC (**Figure 1—figure supplement 1B,C**, **Figure 4—figure supplement 2A**). For the ones that do have some brain expression, optogenetic activation in decapitated flies still produces leg extension phenotypes (**Figures 1E-H**, **Figure 1—figure supplement 1A**).

Conversely, silencing these neurons locks the front legs in flexion (**Figures 1E** and **1E'**, **Figure 1—Video 1**). Thus, activation or silencing of inhibitory neurons interferes with the alternation of flexion and extension required for dust removal and reduces grooming.

Spatial Mapping and Connectivity Patterns of Premotor 13A Neurons in Leg Motor Control

The *Drosophila* nervous system develops from neuroblasts, each of which produces two hemilineages. These lineages arise from neuroblasts that divide briefly embryonically to produce primary neurons and later generate secondary neurons post-embryonically^{40,41,81,82,83,84,85}. We focus on the 13A and 13B hemilineages. Using serial section electron microscopy dataset¹⁵, we identified 62 13A neurons and 64 13B neurons in the right front leg neuromere (T1-R) of VNC. Upon entering the neuropil, 13A bundle divides into three sub-bundles, with five large neurons having extensive arbors, mostly in the ventral first sub-bundle of T1-R (**Figure 2—video 1**). Based on size, location, and connections, we hypothesize these represent early-born primary 13A neurons. We also identified them in the left hemisegment distributed across three sub-bundles.

Lineages contain neurons with distinct shapes and functions. We used NBLAST⁸⁶, a computational tool designed to group neurons based on morphological similarities, to categorize 13A and 13B neurons into distinct clusters (**Figure 2A**, **Figure 2—figure supplement 3**, **Figure 2—video 2**). Connectivity patterns were analyzed using automatic synapse detection algorithms¹⁶. While 13A neurons have many post-synaptic partners, their primary targets are MNs (**Figure 2—figure supplement 4**). Each morphological 13A cluster connects to distinct sets of MNs. Clustering by MN connections correlated with NBLAST morphological clusters, as demonstrated by a cosine similarity matrix (**Figure 2** and **Figure 2—figure supplement 1**). While morphological and connectivity based clusters align, exceptions include cluster 9 and the 13A-3g, which have more diverse connections (**Figure 2B** and **Figure 2—figure supplement 1**).

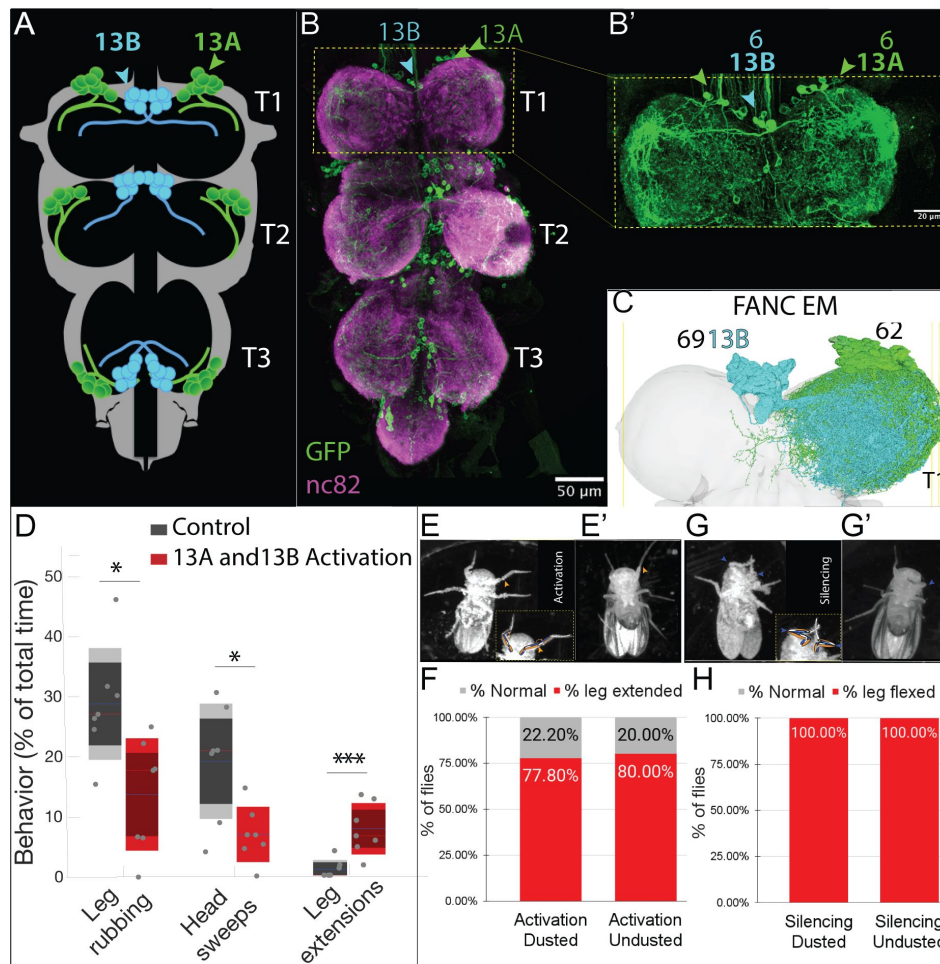


Figure 1.

Anatomical Distribution and Behavioral Contributions of 13A and 13B Hemilineages

(A) Schematic showing segmental distribution of 13A (green) and 13B (cyan) neurons across pro-, meta-, and meso-thoracic segments (T1, T2, T3) of VNC. (B) Confocal image: Six GABAergic 13A neurons (green arrowheads) and six 13B neurons (cyan arrowheads) in each VNC hemisegment, labeled with GFP (green) driven by *R35G04-GAL4-DBD*, *GAD-GAL4-AD*. Neuropil in magenta (nc82). Panel B' provides a zoomed-in view of T1 region. (C) EM reconstructions: 62 13A neurons (green) and 64 13B neurons (cyan) in right T1. Ventral side up. (D) Continuous activation of 13A and 13B neurons labeled by *R35G04-GAL4-DBD*, *GAD-GAL4-AD* in dusted flies reduces front leg rubbing and head sweeps, and induces unusual leg extensions. Control: *AD-DBD EMPTY SPLIT > UAS CsChrimson* (gray). Experiment: *R35G04-GAL4-DBD*, *GAD-GAL4-AD > UAS CsChrimson* (red). Box plots indicate the percentage of time dusted fly engaged in a given behavior over a 4-minute assay ($n = 7$). The solid blue line marks the mean, dark shading the 95% confidence interval, red dashed line the median, and light shading ± 1 standard deviation. *** $P \leq 0.001$, * $P \leq 0.05$. (E-F) Continuous activation of 13A and 13B neuron subsets induces front leg extension in headless flies. (E, E') Representative video frames showing headless flies (dusted and undusted) with extended front legs (orange arrowhead) following continuous optogenetic activation of neurons labeled with *R35G04-GAL4-DBD*, *GAD-GAL4-AD > UAS-CsChrimson*. Dashed box in E highlights the front legs; schematic illustrates the extended posture. (F) Quantification of leg extension phenotypes in dusted and undusted headless flies. Bar plots show the percentage of flies displaying leg extension (red) or a normal posture (gray). Percentages are calculated as the number of flies showing each posture divided by the total number of flies per condition. Dusted: $n = 9$; undusted: $n = 5$. (G-H) Silencing 13A and 13B neuron subsets locks front legs in flexion in headless flies. (G, G') Representative video frames showing dusted and undusted headless flies with sustained front leg flexion following silencing of neurons labeled with *R35G04-GAL4-DBD*, *GAD-GAL4-AD > UAS-TNTe*. Blue arrowheads indicate the flexed posture. (H) Quantification of leg flexion phenotypes in dusted and undusted headless flies. Bar plots show the percentage of flies displaying sustained flexion (red). All flies (100%) in both dusted ($n = 13$) and undusted ($n = 9$) conditions showed the phenotype. Also see **Figure 1—Video 1**.

B6, H1-H9 [↗](#)). Our initial analysis used 13A neurons in the right front leg neuromere (T1R). Comparison to a similar set on the left²¹ [↗](#) revealed similar numbers of neurons and cluster divisions (Figure 2—figure supplement 2 [↗](#)).

Anatomical features of 13A types suggest possible functional organization. Their dendrites occupy specific regions of VNC, suggesting common pre-synaptic inputs. Axons of 13A neurons overlap with MN dendrites (**Figures 2C** [↗](#) and **2D** [↗](#), **Figure 2—video 3** [↗](#)), which are spatially segregated by the leg muscles they innervate, forming a myotopic map^{18,19} [↗](#). Cosine similarity between morphological clusters and MN targets indicates that the spatial position of 13A neurons predicts which MN groups they connect to. Although 13A neurons respect the myotopic organization of motor neurons, many of their connections span multiple MN groups. This shows that the 13A spatial map could reflect premotor synergies (coordinated multi-joint control) rather than a strict one-to-one mapping between neurons and individual muscles.

Some 13A neurons connect to multiple MNs across various leg segments; others target only a few. We classify these as ‘generalists’ and ‘specialists’. We propose that the broadly-projecting generalists are early-born primary neurons and that the specialists that target fewer MNs are later-born secondary neurons. This is consistent with the known developmental sequence of hemilineages, where early-born primary neurons typically acquire larger arbors and integrate across broader premotor and motor targets, whereas later-born secondary neurons often have more spatially restricted projections and specialized roles^{18,19,81,82,85} [↗](#). Our morphological clustering supports this idea: generalist 13As have extensive axonal arbors targeting motor neurons that control multiple leg segments, whereas specialist neurons are more narrowly tuned, connecting to a few MN targets within a segment. Thus, both their morphology and connectivity patterns align with the expectation from birth-order-dependent diversification within hemilineages. Four primary 13A neurons (13A-10f-α, -9d-γ, -10g-β, and -10e-δ) are generalists (Figure 2—figure supplement 1 I5-I8 and I9 [↗](#)). Secondary neurons in clusters 1, 2, 4, 5, and 7 neurons are specialists, while clusters 6, 8 and 10 are generalists. Clusters 3 and 9 contain a mix. Specialist neurons tend to target MNs in a more restricted, segment-specific manner, reflecting a myotopic map, whereas generalist inhibitory neurons target specific MN groups across multiple leg segments, reflecting premotor synergies—coordinated group of muscles that work together. These overlapping patterns suggest that 13As form a spatial map organized by muscle synergies, rather than a strict one-to-one myotopic map. Together, these findings indicate that 13A neurons constitute a spatially organized premotor map, where morphology and connectivity jointly predict the recruitment of both broad, multi-joint synergies and restricted, joint-specific motor outputs.

Connectivity Motifs for Coordinated Control

The adult *Drosophila* leg consists of five joints, named according to the segments they connect: bodywall/thoraco-coxal (Th-C), coxa–trochanter (C-Tr), trochanter–femur (Tr-F; fused in the adult), femur–tibia (F-Ti), tibia–tarsus (Ti-Ta). Each joint is powered by opposing flexor and extensor muscles that transmit force through tendons¹⁵ [↗](#). The proximal joints, Th-C and C-Tr, mediate leg protraction–retraction and elevation–depression, respectively⁵ [↗](#). The medial joint, F-Ti, acts as the principal flexion–extension hinge and is controlled by large tibia extensor motor neurons and flexor motor neurons^{15,16,18,19,21} [↗](#). By contrast, distal joints such as Ti-Ta and the tarsomeres contribute to fine adjustments, grasping, and substrate attachment¹⁶ [↗](#).

We analyzed the connectivity of 13A/B neurons synapsing onto MNs of the medial (F-Ti) joint (**Figure 3** [↗](#)). Interconnections between 13A neuron types suggest a role in generating flexor–extensor antagonism. 13A neurons synapsing onto extensor MNs also inhibit 13A neurons targeting flexors, and vice versa (**Figures 3A2** [↗](#), **3A3** [↗](#), **3B** [↗](#)). These redundant circuits could ensure that at a given time point either extensor or flexor is active.

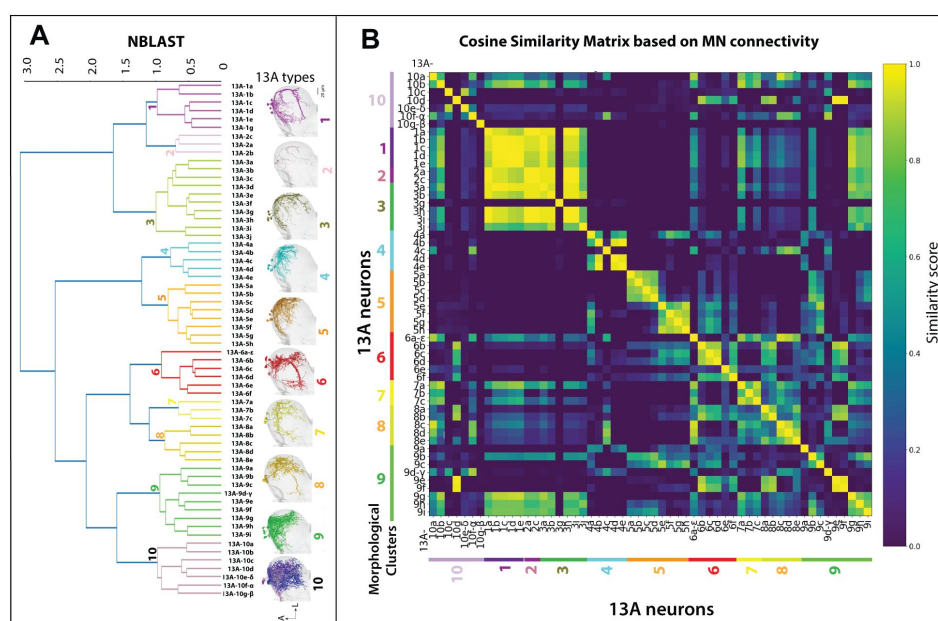


Figure 2.

Spatial Map of Premotor 13A Neurons Correlates With Their Connections to Motor Neurons (MNs)

(A) Hierarchical Clustering of 13A Hemilineage. Clustering of 13A neuron types in the right T1 segment was performed using NBLAST, resulting in identification of 10 morphological groups or *Clusters*. EM reconstructions of distinct 13A clusters are shown. Neurons are named based on morphological clustering. For example, all neurons in the 13A-3 cluster have similar morphology, with 10 neurons labeled as 13A-3 (a-j) (olive). Images of each 13A neuron are shown in *Figure 2—Figure Supplement 1*. Also see *Figure 2—Video 2*. A= anterior, L= Lateral. Ventral side is up. **(B) Cosine similarity graph showing pairwise similarity between 13A neurons based on their MN connectivity patterns.** 13A neurons are organized based on anatomical clusters obtained with NBLAST as described above. It depicts a correlation between anatomy of 13A neurons and their connections to MNs. For example, 13A-1a, -1b, -1c, -1d (cluster 1) connect to same set of MNs, therefore have high cosine similarity with each other (as seen across the diagonal). Graph also gives insights into groups of 13As that control similar muscles. For example, cluster 1 neurons have high cosine similarity with cluster 3 13A neurons (while, 3g neuron is an exception).

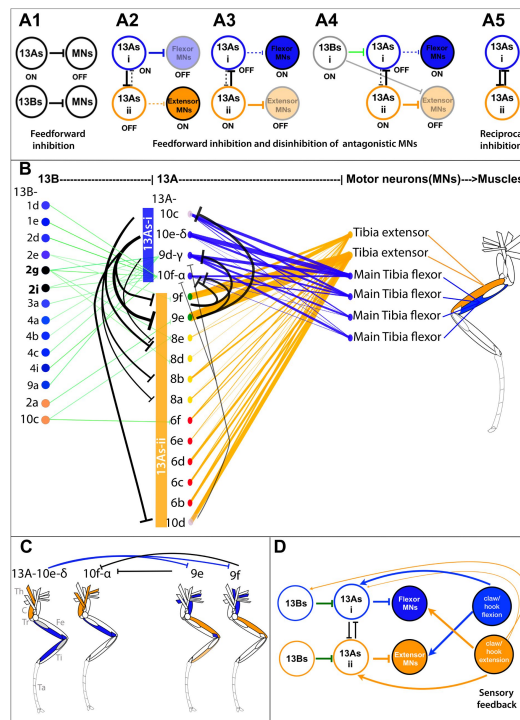


Figure 3.

Inhibitory Circuitry for Antagonistic Muscle Control

(A) Schematic of inhibitory circuit motifs (A1) Feedforward inhibition by 13A/B neurons. **(A2)** Flexor inhibition and extensor disinhibition: 13As-i inhibit flexor MNs and disinhibit extensor MNs by inhibiting 13As-ii. **(A3)** Extensor inhibition and flexor disinhibition: 13As-ii inhibit extensor MNs and disinhibit flexor MNs by inhibiting 13As-i. **(A4)** 13B mediated disinhibition: 13Bs disinhibit MNs by targeting premotor 13As, while some also directly inhibit antagonistic MNs. **(A5)** Reciprocal inhibition among 13A groups that inhibit antagonistic MNs may induce flexor-extensor alternation. **(B) Connectivity matrix:** Inhibitory connections regulating antagonistic MNs of the medial joint. Leg schematic shows tibia extensor (orange) and flexor (blue) muscles, innervated by respective MNs. Flexor-inhibiting 13A neurons (13As-i) in blue, and extensor inhibiting 13As (13As-ii) in orange. The thickness of edges between nodes is determined by number of synapses. Node colors were assigned based on the type of neurons, with specific colors denoting different subtypes of 13A/B neurons and MNs. **Feedforward inhibition:** Primary neurons (13A-10f-α, 9d-y, and 10e-δ) and 13A-10c (13As-i) connect to tibia flexor MNs (blue edges), making a total of 85, 219, 155, 157 synapses, respectively. Twelve secondary 13As ii inhibit tibia extensor MNs (orange edges), with strong connections from 13A-9f, -9e, and -10d totaling 188, 275, 155 synapses, respectively. **Reciprocal inhibition:** Three neurons from 13As-i inhibit six from 13As ii, with 13A-10e-δ connecting to 13A-9f (19 synapses), -9e (31), and -10d (14). 13A-10c connects to 13A-8a (6), -8b (12), and -8e (5). 13A-9d-y connects to 13A-8e (8). Conversely, three from 13As-ii inhibit two neurons from 13As i, with 13A-9f connecting to 13A-10f-α (25) and -9d-y (6), and 13A-10d connecting to 13A-10f-α (8), -9d-y (7), and -10e-δ (15). 13A-9e connects to 13A-10f-α (21) and -10c (47) (black edges). **Disinhibition by 13B neurons:** 13B connects to 13As-i (13A-10f-α and -9d-y) (totaling 78 and 50 synapses) (green edges), disinhibiting flexor MNs. 13B-2g and -2i also directly inhibit tibia extensor MNs. **(C) Reciprocal inhibition for multi-joint coordination:** Primary 13As (10e-δ and 10f-α) target a combination of proximal (sternotrochanter, tergotrochanter, trochanter extensor, tergotrochanter), medial (tibia flexor), and distal (tarsus depressor) MNs, while secondary 13As (9e and 9f) target antagonist MNs including sternal posterior rotator, pleural remotor abductor, and tibia extensor. Reciprocal connections between them indicate that generalist 13As coordinate multi-joint muscle synergies through inhibition of antagonistic motor groups. Leg schematic shows the muscles innervated by the corresponding MNs in various leg segments (Th = thorax, C = coxa, Tr = trochanter, Fe = femur, Ti = tibia, Ta = tarsus). (Data for 13A-MN connections are shown in [Figure 2—figure supplement 1 19, 16, 17, H9, H4, and H5](#); 13A-13A connections shown in [Figure 3—figure supplement 1C](#)). **(D) Proprioceptive feedback:** Sensory feedback from proprioceptors onto reciprocally connected 13As could turn off corresponding MNs and activate antagonistic MNs. Flexion-sensing proprioceptors target extensor MNs and 13As-i that inhibit tibia flexor MNs. Extension-sensing proprioceptors target tibia flexor MNs and two 13As (13As-ii) that inhibit extensor MNs. Claw extension neurons also connect to 13A-δ. One 13B that disinhibits flexor MNs also receives connection from extension-sensing proprioceptors. Also see [Figure 3—figure supplement 3](#).

These 13A groups are also reciprocally connected to each other (**Figures 3A4**, **3A5**, **3B**) providing a mechanism for alternation between flexion and extension over time.

This organizational motif applies to multiple joints within a leg as reciprocal connections between generalist 13A neurons suggest a role in coordinating multi-joint movements in synergy (**Figure 3C**).

We did not find any correlation between the morphology of premotor 13B and motor connections, but there is a topographically restricted output based on their 13A premotor targets (**Figure 2—figure supplement 1** and **Figure 3—figure supplement 1B**). 13B neurons connect to 13A neurons targeting either flexor or extensor MNs (**Figure 3—figure supplement 1B** and **Figure 3—figure supplement 2**). Two specific 13B neurons inhibit both extensor MNs and disinhibit flexor MNs, playing a dual role. Twenty-four 13B neurons from clusters 1 to 4 target 13A neurons.

Interconnections between 13A and 13B neurons reveal additional inhibitory motifs that could mediate movement of other joints or multiple joints synergistically. 13B neurons disinhibit MNs by inhibiting premotor 13Bs or 13As. For example, 13B-4h inhibits 13B-2i, a generalist premotor neuron targeting proximal flexor and medial extensor MNs (**Figure 3—figure supplement 1A, E, F**), while preventing disinhibition of antagonistic MNs (**Figures 3B** and **Figure 3—figure supplement 1B**). Similarly, 13As could disinhibit antagonistic MNs by inhibiting premotor 13Bs. For example, primary 13A-10g- β connects to 16 13B neurons, disinhibiting proximal extensor MNs while inhibiting proximal/medial flexors (**Figure 3—figure supplement 1A, D, F**).

Together, inhibitory interconnections among 13A neurons, and between 13A and 13B neurons, may coordinate alternating activity in antagonistic muscle groups.

Proprioceptive Feedback to 13A Neurons

We identified connections from position-sensing proprioceptors to primary 13A neurons. These could provide sensory feedback. Claw neurons detect position, while hook sense movement direction^{87, 88, 89}. We examined reconstruction of proprioceptive neurons²⁰ and found multiple connections from flexion-sensing claw and hook neurons onto the main neurite of 13A-10f- α , which targets tibia flexor MNs (**Figures 3D** and **Figure 3—figure supplement 3**). Similar connections were observed onto 13A-10e- δ , -9d- γ . Recent connectome analysis showed that flexion sensing proprioceptors send direct excitatory feedback to tibia extensor MNs and indirect inhibitory feedback to flexor MNs⁴⁷. Thus, flexion-sensing proprioceptors could activate primary 13A neurons to inhibit tibia flexor MNs, while directly activating extensor MNs. Similarly, claw extension neurons connect to two 13A neurons that target tibia extensor MNs, while directly connecting to flexor MNs. Since these two groups of 13A neurons receive proprioceptive feedback and reciprocally inhibit each other, they could drive flexion-extension alternation.

Behavioral Evidence for Muscle Synergies During Grooming

Dusted flies use their legs to perform precise grooming actions, involving repeating patterns of body sweeps and leg rubs^{6, 7}. Quantifying these movements using machine vision methods (DeepLabCut)⁹⁰ reveals synchronized changes in angular velocity across multiple leg joints (**Figure 4A**). During leg rubbing, proximal and medial joints within a given leg move predominantly in sync, as indicated by a minimal lag in their angular velocities (**Figure 4A'**), though they occasionally move asynchronously during head sweeps (**Figure 4A''**). This coordination shows the presence of muscle synergies, and generalist premotor interneuron connectivity could be how these synergies are implemented.

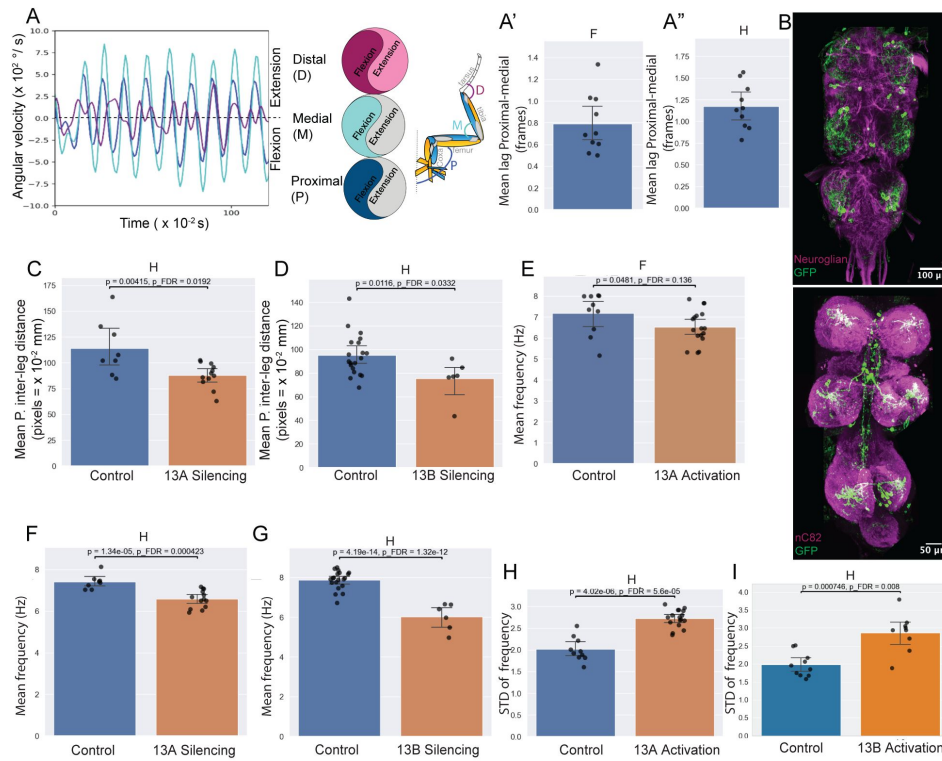


Figure 4.

13A and 13B Neurons Are Required for Leg Coordination During Grooming

(A-A'') **Intra-joint coordination and muscle synergies.** (A) Angular velocities of proximal (P, blue), and medial (M, cyan) joints predominantly move synchronously, while distal (D, purple) can move in or out of phase during leg rubbing. The schematic (right) indicates the corresponding joint angles. (A'-A'') The proximal and medial joint movements within a leg occur effectively in phase, with a mean lag of ~0.8 frames (8 ms) during leg rubbing (A') and during head grooming sweeps (A''). Bar plots show the lag; each dot indicates one animal. Frame = 10ms. **(B) Neuronal labeling of 13A and 13B neurons.** Top: Confocal image of six Dbx positive 13A neurons per hemisegment labeled by GFP using *R35G04-GAL4-DBD, Dbx-GAL4-AD* in VNC. Neuroglial (magenta) labels axon bundles. Bottom: Confocal image of three 13B neurons per hemisegment labeled by GFP using *R11B07-GAL4-DBD, GAD-GAL4-AD*. Nc82 (magenta) labels neuropil. **(C-I) Effects of neuronal activity manipulation in dusted flies.** Silencing and activation of 13A neurons in dusted flies using *R35G04-GAL4-DBD, Dbx-GAL4-AD* with *UAS Kir* or *UAS CsChrimson*, respectively (n=12 silencing, n=19 activation). Control: *AD-GAL4-DBD EMPTY SPLIT* with *UAS Kir* or *UAS CsChrimson*. For 13B neurons, *R11B07-GAL4-DBD, GAD-GAL4-AD* with *UAS GtACR1*, or *UAS CsChrimson* respectively (n=7 silencing, n=9 activation); control: *AD-GAL4-DBD EMPTY SPLIT* with *UAS GtACR1* or *UAS CsChrimson*. Each panel compares control (blue) and experimental (orange) groups. Each dot represents the mean feature value for a single fly. Bars indicate the group mean, and whiskers represent the 95% confidence interval of the group mean. P-values (raw and false discovery rate [FDR]-corrected) are shown above each panel. **(C-D) Proximal inter-leg distance:** Silencing of 13A (C) or 13B (D) neurons during head grooming reduces the distance between the femur-tibia joints of the left and right front legs. **(E-I) Frequency modulation:** Silencing 13A or 13B neurons reduces mean frequency of proximal joint oscillations in dusted flies. (F, G). Activation of 13A neurons reduced frequency, although this change did not survive FDR correction. However, continuous activation of 13A and 13B neurons increased variability in frequency. (H, I). Mean of the per-animal standard deviation (STD) that reflects variability or spread of data is shown.

13A Neurons Affect Limb Coordination During Grooming

Half of the 13A population expresses a transcriptional factor, *Dbx*⁴⁰. We used Split GAL4 combinations to target smaller subsets, intersecting with a GAD line and *Dbx* to manipulate a small subset of 13A neurons.

These 13A neurons (**Figure 4B** and **Figure 1—figure supplement 1B**) target MNs controlling multiple joints, including proximal (Th-C) (Sternotrochanter extensor, tergotrochanter, tergoplural promotor body wall muscles, and trochanter extensor MNs), medial (F-Ti) joint extensor/flexor MNs, and distal (Ti-Ta) joint tarsus extensor MN (**Figure 4—figure supplement 1A1-1A3**). Based on their connectivity, we hypothesized that silencing 13A neurons, and thereby removing inhibition from these MNs, would lead to exaggerated or mistimed movements at proximal and medial joints and improper positioning of the distal tarsus. Specifically, proximal joints may show excessive retraction and elevation, medial joints may display uncoordinated flexion/extension, and distal tarsus movements may become poorly targeted or overshoot, resulting in reduced inter-leg spacing and inefficient, uncoordinated grooming strokes. Conversely, activating 13As would suppress MN activity, reducing leg rotation and joint movements, limiting tibia bending and tarsus contact, and thereby impairing coordinated grooming.

To test function, we manipulated 13A neuron activity in intact, behaving animals, where motor output and sensory feedback interact continuously. This allowed us to probe their role within the intact sensorimotor system. Given their connectivity with both motor neurons and proprioceptive inputs, 13A neurons likely contribute to movement generation and its modulation by feedback. Therefore, the behavioral outcomes observed in our assays reflect their integrated role in motor control.

We measured changes in extension and flexion with joint positions, angles and inter-leg distances. Activating or silencing six 13A neurons reduced total grooming time and disrupted joint positions in dusted flies (**Figure 4—figure supplement 1E'-1F'**). Silencing 13A neurons reduced mean distance between F-Ti joints of the front legs (**Figure 4C**) and decreased the frequency of extension-flexion cycles (**Figure 4F**) indicating that 13A activity is required to sustain rhythmic motor output. Continuous activation of these neurons also showed a trend toward reduced frequency (though not statistically significant after correction) and increased variance suggesting that persistent activity makes the leg movements slower and more variable (**Figures 4H**).

To test whether these effects were specific to grooming behavior, we also examined walking bouts in these dusted flies. Silencing 13A neurons did not affect the frequency of extension-flexion cycles during walking in dusted flies (**Figure 4—figure supplement 1G**). However, two spatial features were altered: the distance between the tarsal tips of the front legs and the distance between the tibia-tarsus joints were both reduced, indicating subtle effects on front limb posture during walking (**Figure 4—figure supplement 1G**).

Silencing specific 13A neurons in dusted flies disrupted both spatial and temporal features of grooming, highlighting their necessity in producing precise grooming actions.

Furthermore, optogenetic manipulation of two different *Dbx*-positive 13A neurons also disrupted grooming, reducing joint position precision and mean frequency, without significantly affecting maximum angular velocity during grooming (**Figure 4—figure supplement 2**). Thus, 13A neurons regulate leg coordination during grooming.

13B Neurons Affect Limb Coordination During Grooming

Our experiments demonstrated that silencing or activating 13B neurons reduced grooming. Connectome data revealed that 13B neurons disinhibit groups of motor neurons. Most 13B neurons appear to act indirectly in the connectome—by contacting inhibitory neurons, potentially producing a net excitatory effect—while a subset makes direct contacts onto motor neurons. We generated a split GAL4 line that labels three 13B neurons (**Figure 4B** [↗](#), **Figure 4—figure supplement 3** [↗](#), and **Figure 1—figure supplement 1D** [↗](#)): two inhibit a primary 13A neuron (13A-9d-γ) which targets proximal extensor and medial flexor MNs (**Figure 4—figure supplement 3C** [↗](#)), and one is premotor, directly inhibiting both proximal and tibia extensor MNs. Together, these 13B neurons could disinhibit proximal extensor and medial flexor MNs while inhibiting medial extensor MNs.

Activating or silencing these three 13B neurons in dusted flies also reduced grooming and resulted in joint positioning defects (**Figure 4D** [↗](#), **Figure 4—figure supplement 3 D', E'** [↗](#)). Silencing 13B neurons decreased proximal inter-leg distance (**Figure 4D** [↗](#)). Continuous activation of 13B neurons often resulted in one leg being locked in flexion while the other leg remained extended perhaps indicating contribution from unknown left right coordination circuits.

Manipulating 13B neuron activity also affects temporal aspects of grooming. Optogenetic silencing of 13B neurons in dusted flies strongly decreased mean frequencies of extension-flexion cycles of all joints (**Figure 4G** [↗](#)), while activation resulted in increased variability (**Figure 4I** [↗](#)). Both silencing or continuous activation decreased total time spent in anterior grooming, although, the bout duration of head sweeps increased upon silencing and that of leg rubs slightly reduced in dusted flies (**Figure 4—figure supplement 3 F-G'** [↗](#)).

Together, 13A and 13B neurons contribute to both spatial and temporal coordination during grooming.

Activation of Inhibitory Neurons Induces Rhythmic Leg Movements

Connectome analysis revealed that inhibitory 13A and 13B neurons frequently synapse onto 13A premotor neurons. Thus, activation of these 13A or presynaptic 13B neurons should inhibit postsynaptic 13A neurons, releasing activity in MNs and promoting movement. Consistent with this connectivity, we observe that optogenetic activation of specific 13A and 13B neurons triggers grooming movements.

We also examined whether the timing of 13A activation could influence frequency of these rhythmic grooming actions. Anterior grooming actions in dusted flies are rhythmic where leg rubbing and body sweeps typically occur at a median frequency of ~7-8 Hz. One complete extension and flexion cycle, representing one sweep or leg rub, lasts ~140 ms, with 70 ms extension and 70 ms flexion phases (**Figures 5B** [↗](#) and **5B'** [↗](#)). Connectivity analysis suggests that specific 13A neurons would be tuned to induce extension and others induce flexion, with reciprocal inhibition potentially generating rhythmicity. We optogenetically activated specific 13A neurons using 70 ms on and off light pulses, to mimic the flexion-extension cycle, in clean flies. This indeed induced grooming (anterior and posterior) and walking (**Figures 5C** [↗](#)–**5D** [↗](#), **Figure 5—Video 1** [↗](#)). The frequency and maximum angular velocity of proximal joint movements during these induced behaviors closely matched those observed in dust-induced grooming (**Figures 5E** [↗](#), **5F** [↗](#)).

To test whether the induced rhythm was locked to the stimulation period, we delivered optogenetic stimulation with equal on/off pulses of 10 ms (50 Hz), 50 ms (10 Hz), 70 ms (~7 Hz), 110 ms (~4.5 Hz), and 120 ms (~4 Hz) and compared the mean frequency of proximal joint cycles

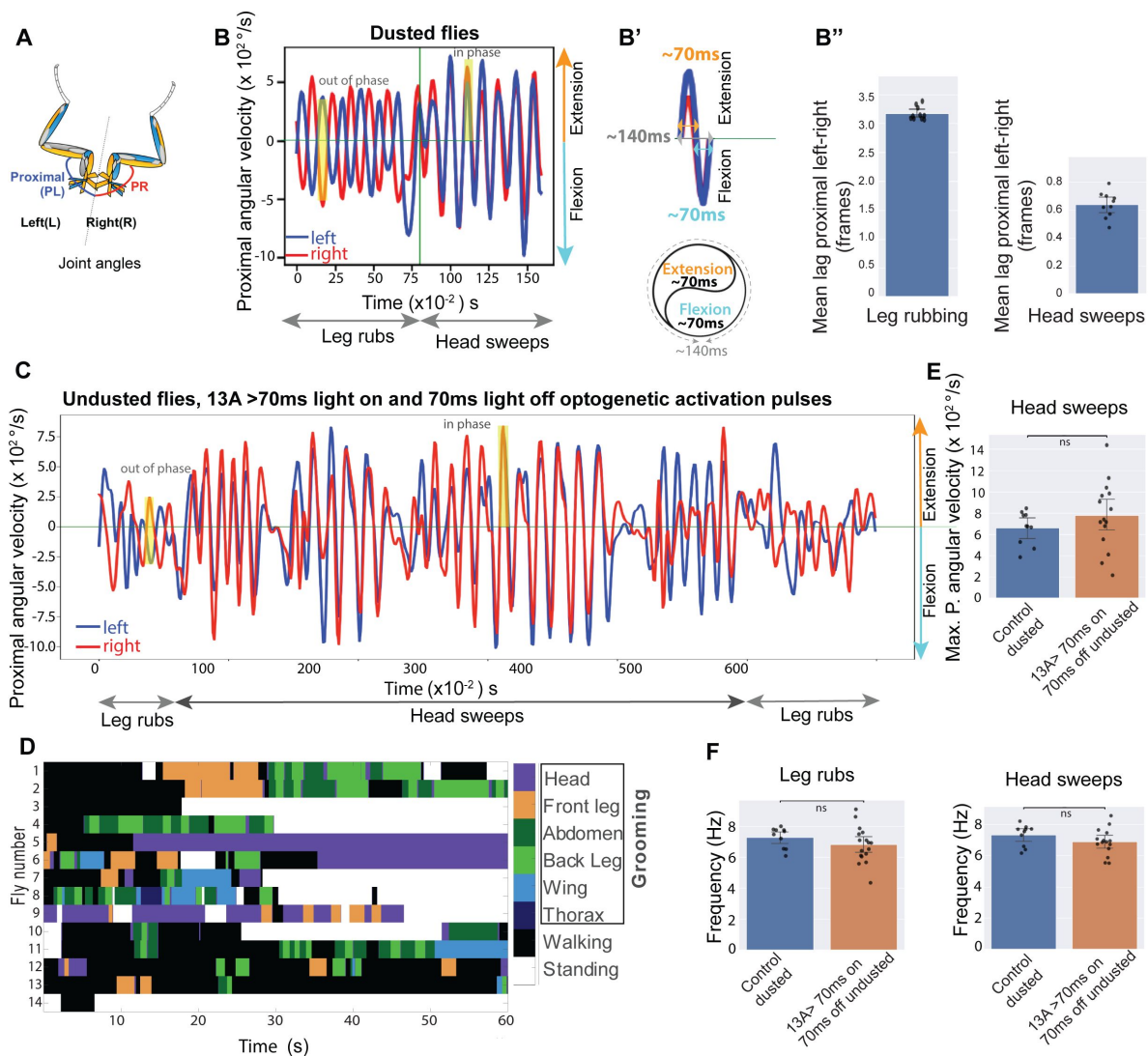


Figure 5.

Pulsed Activation of 13A Neurons Triggers Rhythmic Actions in Clean Undusted Flies

(A) Schematic showing proximal joint angles of left (PL) and right (PR) legs (B) **Left-right coordination and muscle synergies during anterior grooming.** Dusted flies perform alternating leg rubs and head sweeps. Proximal joint angular velocities are shown. PL (blue) and PR (red) joints move anti-phase during leg rubs and in-phase during head sweeps (highlighted yellow box). Positive values indicate extension, and negative indicates flexion. (B') Each flexion and extension cycle lasts ~140 ms, with each phase around 70 ms. (B'') Mean lags between proximal joints of the left and the right legs during leg rubbing and head sweeps. High lag during leg rubbing (left panel) indicates out of phase movement and low lag during heads sweeps (right) indicates in phase movement. Bar plots show the lag; each dot indicates one animal. Frame = 10ms. (C-F) Effect of optogenetic activation using 70ms on and 70ms off pulses in specific 13A neurons (*R35G04-DBD, Dbx-GAL4-AD >UAS CsChrimson*) in undusted flies. (C) Angular velocity of PL and PR leg joints shows anti-phase leg rubs and sustained in-phase head sweeps, with light pulses active from time=0. (D) Behavioral ethogram showing various grooming actions (head, front leg, abdomen, back leg, wing, thorax) and walking triggered by 70ms on and 70ms off pulsed activation of 13A neurons in undusted flies, with light pulses on from time=0. (E) Maximum angular velocity of proximal joints during head sweeps upon pulsed 13A activation in undusted flies is comparable to that observed in dusted flies. (F) The frequency of proximal joint movements during leg rubbing (left) and head sweeps (right) induced by 13A pulsed activation is also similar between dusted and undusted flies. *Control flies: AD-DBD EMPTY SPLIT > UAS-CsChrimson, dusted; Experimental flies: R35G04-DBD, Dbx-GAL4-AD > UAS-CsChrimson, undusted. Light pulses were delivered at 70 ms on / 70 ms off.* Each panel compares control (blue) and experimental (orange) groups. Each dot represents the mean feature value for a single fly. Bars indicate the group mean, and whiskers represent the 95% confidence interval of the group mean. *P*-values (raw and false discovery rate [FDR]-corrected) are shown above each panel.

across conditions. Frequencies did not significantly differ across stimulation paradigms (**Figure 5—figure supplement 1**), suggesting that pulsed activation triggers the circuit's intrinsic rhythm rather than precisely pacing it.

We note that CsChrimson has relatively slow off-kinetics, which may limit the temporal precision of optogenetic control. Nevertheless, across a wide range of stimulation frequencies, optogenetic activation elicited alternating leg movements that were consistent with normal grooming behavior.

This experimental evidence shows that 13A neurons can generate rhythmic movements, reinforcing their role in coordinating grooming behavior.

A Computational Model of Inhibitory Circuits in Coordinating Grooming Actions

The inhibitory circuits connecting to MNs are complex and genetic reagents to target their individual components are limited, so figuring out how each component contributes to leg coordination experimentally is challenging. We developed a neural computational model based on anatomical connectivity to explore potential circuit functions. Several studies have employed (more complex) neural models to show how oscillatory behaviors, such as walking, can emerge without an explicit need for a central controller, e.g. CPG^{4,91}. We modeled groups of functionally related neurons. For example, 13As that are inhibiting each other represent two groups modeled as two nodes – **Figure 6A** shows such a circuit for a single joint. This approach is loosely inspired by Jovanic et al., 2016⁹². Since we are not modeling individual neurons, the network does not involve spiking neurons but rather “rate based” units. The “synaptic weights” of the model network correspond to number of synapses obtained empirically from connectome (**Figure 6A**).

The neural network controls movements of virtual front legs of an agent, where each leg is simplified to a 2D configuration of three segments. A pair of antagonistic “muscles” controls each of the three “joint” angles on each leg. These pairs of muscles determine the angular velocities of each “joint.” Thus each leg receives inputs from six virtual MNs. These MNs receive descending excitatory inputs, and inhibitory inputs from two 13A nodes (inhibiting the MNs of flexors and extensors). The states of the muscles (amounts of extension or flexion) are sensed by sensory neurons (SNs) that provide feedback to the 13As and to MNs. As the legs move, the most distal “joint” removes the virtual dust (**Figure 6I**). Legs must also spend some time in proximity to each other to remove the “dust” from themselves (This constraint forces the legs to coordinate with each other.).

The sensory input to the model circuit is a function of the distribution of “dust” remaining on the “body” - the environment—represented by the green areas in **Figure 6I** and **Figure 6-Video 1**. We use a small recurrent neural network (RNN) that transforms the distribution of the “dust” into excitatory inputs for the 13A network. This simple RNN, consisting of 40 units, is a “black box” used to provide the 13A circuits with excitatory sensory inputs so that the agent can respond to the changing environment. Other inputs to the “black box” include a copy of motor neuron's activation levels (“efference copy”) and the amount of dust accumulated of the legs.

But the 13A circuitry can still produce rhythmic behavior even without those excitatory inputs from the “black box” (or when the inputs are set to a constant value), although the legs become less coordinated (because they are “unaware” of each other's position at any time). Indeed, when we refine the model (with the evolutionary training) without the “black box” (using a constant input of 0.1) the behavior is still rhythmic although somewhat less sustained (**Figure 7**). This confirms that the rhythmic activity and behavior can emerge from the modeled pre-motor circuitry itself, without a rhythmic input.

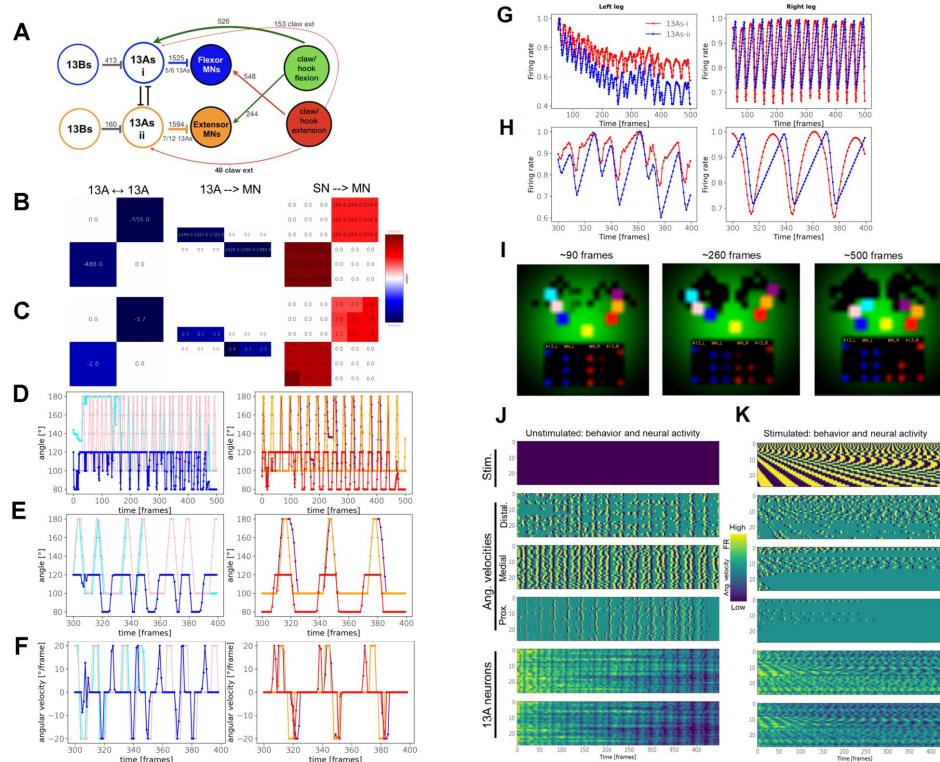


Figure 6.

Modeling the 13A Circuits.

(A) Circuit diagram showing inhibitory circuits and synaptic weights based on connectome. (B) Adjacency matrices from the empirically estimated weights, indicated in the simplified circuit diagram in (A). The 13B neurons in this model do not connect to each other, receive excitatory input from the black box, and only project to the 13As (inhibitory). Their weight matrix, with only two values, is not shown. Excitatory and inhibitory connections are shown in red and blue, respectively. (C) Adjacency matrices of the model circuits same as in (B) but after fine-tuning. The three “joint” angles of the left leg (left) and the right leg (right) as they change over the time of one episode (500 frames). Colors indicate “joints” as follows: distal (cyan), medial (pink), and proximal (blue) for the left leg. Right leg: distal (purple), medial (orange), and proximal (red). (D) Same as (D) but zoomed-in to between 300-400 frames. (E) Same as (E) but showing angular velocities [°/frame]. (F) Firing rates (activity levels) of the two 13A neurons (red and blue) over one episode (500 frames), for both legs (left, right). (G) Same as (G) but zoomed-in to between 300-400 frames. (H) Video frames from the beginning, middle, and end of a video of one episode. Left leg is represented by three “joints”: distal (cyan), medial (pink), and proximal (blue). Right leg: distal (purple), medial (orange), and proximal (red). The legs originate from the “base” (yellow). As legs move over the “body” (the environment – dust is represented as the green Gaussian distribution), the dust (green) is getting removed (black background). The bottom of each movie frame shows the activity of the two left 13A nodes and six left MNs (blue). The right leg nodes are shown in red, on the right side. Brightness of the nodes indicates the activity level. See [Figure 6-Video 1](#). (I) The dynamics of angular velocities of the left leg’s “joints”, and left 13A activation levels, over 100 episodes (500 frames each), when no stimulus is given (indicated by empty matrix on the top). Each row of each matrix is one episode. (The simulation started running for 50 frames before Time=0 but it starts with very high peaks which were not plotted here for better visualization.) (J) Same as in J, but stimulation with pulses of varying durations is given. Top row of each matrix: pulse duration=2 frames; bottom (100th) row of each matrix: pulse duration=100 frames. The pulse stimulation is indicated in the top matrix.

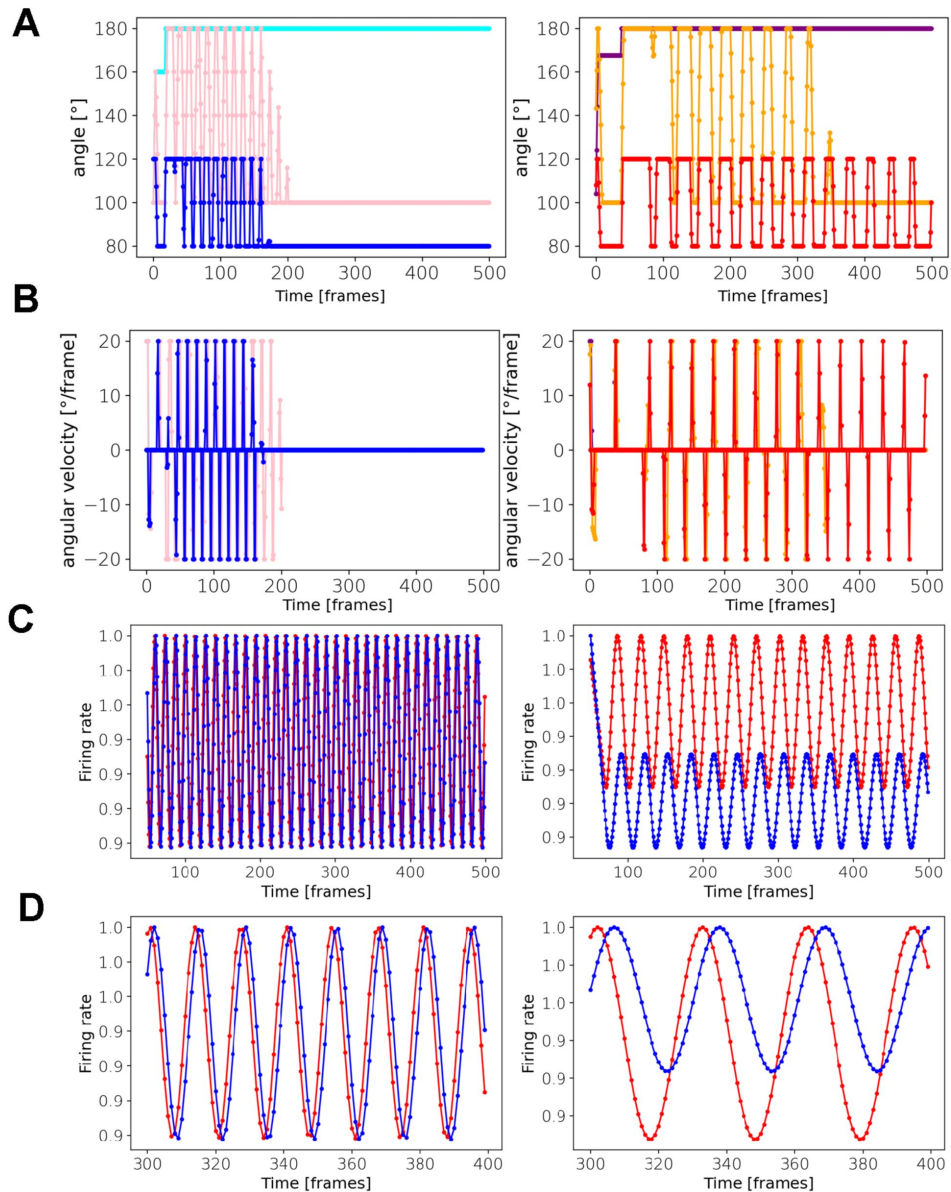


Figure 7.

The Modeled 13A Circuits Can Produce Rhythmic Behavior and Activity Without Rhythmic External Input

(A) The three “joint” angles of the left leg (left) and the right leg (right) as they change over the time of one episode (500 frames). Colors indicate “joints” in the as follows: distal (cyan), medial (pink), and proximal (blue). Right leg: distal (purple), medial (orange), and proximal (red). (B) Same as (A) but showing angular velocities [°/frame]. (C) Firing rates (activity levels) of the two 13A neurons (red and blue) over one episode (500 frames), for both legs (left, right). (D) Same as (C) but zoomed-in to between 300–400 frames.

We also added 13B nodes, as shown in **Figure 6A** [↗](#). These nodes receive inputs from the same ‘black box’ as the 13As.

Figure 6B [↗](#) shows three types of adjacency matrices obtained from the connectome. The adjacency matrices of the model are scaled versions of these three matrices. (We assume that the ratio between the weights in each adjacency matrix, rather than the absolute numeric values in **Fig 6B** [↗](#), reflects the connectivity.) Each adjacency matrix is scaled separately (see Methods, the fine-tuning of the synaptic weights section), preserving the ratios of synaptic weights between the same types of neurons. When we run the model with the exact values of the adjacency matrices, we do not get any behavior. So, we instead allowed the model’s parameters to deviate around their empirically obtained values. We can do this because our empirically derived weights are not exactly represented by the weights of the modeled network. Thus, we “fine-tune” the weights, subject to constraints based on empirical values: the signs of the fine-tuned weights must remain the same as the empirical synapses (e.g. inhibitory neurons remain inhibitory), and their magnitudes have the upper and lower bounds of 20% above and below the empirical weights. No synapse can be removed from, or added to, the model by the fine-tuning process. The fine-tuning procedure was originally accomplished by genetic algorithms (GA) library PyGAD (<https://pygad.readthedocs.io/en/latest/index.html> [↗](#)), and is currently done by our own genetic algorithm (<https://github.com/PrimoRavbar/Inhibitory-circuits> [↗](#)). The fitness function is defined as the total amount of virtual dust that the model agent removes across several episodes of grooming. The genomes contain the parameters (synaptic weights and thresholds) and hyper-parameters (see Methods). The fine-tuned weight matrices and the original ones are shown in **Figures 6B** [↗](#)–**6C** [↗](#). Note how the ratios between synaptic weights are largely preserved.

After the fine-tuning, we analyzed the activity of the modeled 13A neural circuits, and the behavior it produced. **Figure 6I** [↗](#) shows the first, middle, and last frame of a movie. The agent succeeded in removing most of the dust (the green pixels). **Figures 6D** [↗](#)–**6F** [↗](#) shows the angles and angular velocities of the three “joints” of each front leg, and **Figures 6G** [↗](#), **6H** [↗](#) show the corresponding neural activity of the 13A neurons. Notice the periodic patterns in both, the motor output and the firing rates of the 13A nodes.

Next, we inquired how the model responds to perturbations analogous to the experimental activations of the 13A neurons. **Figure 6J** [↗](#) shows the dynamics of left leg joint angles in 100 renditions, when no stimulus is applied. Notice the regular periodicity of these dynamics. As we vary the length of activation pulses (**Figure 6K** [↗](#)), behavior, as reflected in angle dynamics, becomes distorted. (Legs also lose coordination and consequently less dust is removed.) These distortions also involve higher frequency of movements (see angular velocities in **Figure 6K** [↗](#)).

We also tried removing individual synaptic connections: removal of either one of the reciprocally inhibiting connections between 13As of a leg completely paralyzes it. Removing all ‘proprioceptive’ feedback from SNs to MNs does not stop the execution of the periodic movement, but it slows it down (**Figure 6—figure supplement 1** [↗](#)). Obliterating all 13A → MN synapses, not surprisingly, completely paralyzes the leg. And, when we remove just the 13A-i-MN connections (which control the flexors of the right leg) we likewise get a complete paralysis of the leg. However, removing of the 13A-ii-MN (which control the extensors of the right leg) has only modest effect on leg movements. So, we need the 13A-i neurons to inhibit the flexors (via motor neurons), but not extensors, in order to obtain rhythmic movements. Thus, our computational model confirms that rhythmic movements could be produced by inhibitory 13A circuits (even without external sensory inputs from the dust, or patterned descending input). By replicating leg movements based on real anatomical connectivity, we investigated the potential functional roles of specific circuit components. This platform enhances our understanding of inhibitory circuits in leg coordination during grooming, providing a foundation for generating informed hypotheses in future experimental studies.

Discussion

Inhibitory Circuit Motifs

Using VNC EM connectome data^{16,20}, we identified various circuit motifs formed by 13A/B inhibitory neurons that contribute to motor control.

Feed-forward inhibition

Generalist 13A neurons synapse onto multiple MNs influencing broad movements like whole leg extension, while specialist 13A neurons could refine joint-specific movements (Figure 2—figure supplement 1).

Disinhibition

13A neurons targeting extensor MNs connect to 13As targeting flexor MNs (Figure 3), enabling flexor activation when extensors are inhibited. Generalist mediated disinhibition coordinates muscle synergies across joints, promoting leg extension or flexion. Some 13B neurons provide direct inhibition (Figure 3A), while others have an indirect effect by disinhibition of motor pools. This disinhibition motif, similar to those observed in motor systems for sequence selection^{92,93,94,95,96,97,98} and flight regulation¹⁰⁰, may prime motor responses by holding them at the ready, to be released when inhibition is removed. Moreover, alternating inhibition and disinhibition of antagonistic motor neuron pools by premotor CPG networks has been observed previously in stick insects^{25,28,79}. Disinhibitory motifs are also present in *Drosophila* larvae: interlinked inhibitory interneurons implement lateral and feedback disinhibition to guide sensorimotor decisions⁹², and selective inhibition by premotor interneurons coordinates the sequential activation of motor neurons⁹⁹. Our connectome analysis extends these concepts by identifying inhibitory 13A/B motifs that could implement such coordination in the adult *Drosophila* VNC to control leg movements.

Reciprocal inhibition

13A neurons inhibiting flexors and extensors within a leg are reciprocally connected (Figure 3). This circuit resembles r1a-inhibitory neurons that are reciprocally connected circuits involved in vertebrate locomotor rhythm generation^{100,101,102,103,104,105,106}. Reciprocal inhibition is also a well-established mechanism in insect motor systems: in stick insects, pilocarpine-induced rhythmic activity in deafferented nerve cords reveals strictly alternating activity between antagonistic motor neurons¹⁰⁷, and intracellular recordings show cyclic hyperpolarizing inputs to flexor and extensor motor neurons in antiphase²⁵. Similarly, locust thoracic ganglia exposed to pilocarpine exhibit alternating phases of antagonistic motor activity, indicating that reciprocal inhibitory motifs are a conserved solution for generating alternation¹⁰⁸. Our connectomic analysis extends these classic findings by demonstrating reciprocal inhibition between inhibitory premotor 13A neurons themselves, rather than solely between antagonistic motor neurons. This additional layer of reciprocal connectivity suggests that 13As may act as a rhythm-generating kernel that shapes the timing of motor output and coordinates antagonistic activation.

Redundant inhibition

Inhibitory neurons target both MNs and their excitatory pre-synaptic partners, creating a parallel pathway that could modulate motor output through direct and indirect inhibition. For example, 13A-10f-α connects to both tibial flexor MNs and excitatory premotor neurons (20A/3A) that activate flexor MNs (Figure 2—figure supplement 4), preventing their activation by two

parallel pathways. Similarly, the 13B-4i neuron connects to 13A neurons inhibiting flexor MNs, leading to disinhibition (Figure 3—figure supplement 2 [↗](#)), and to excitatory neurons presynaptic to 13A neurons, thereby both removing inhibition from flexor MNs and limiting the excitatory drive onto 13A neurons.

Proprioceptive feedback onto inhibitory circuits

Position-sensing proprioceptors connect to 13As (Figures 3D [↗](#) and Figure 3—figure supplement 3 [↗](#)), which inhibit flexion and disinhibit extension and vice versa, complementing reciprocal inhibition to generate alternation. Movement-sensing proprioceptors synapse onto 13A neurons, but are suppressed during walking and grooming¹⁰⁹ [↗](#). Sherrington's 1910 proposal, supported by spinal cat studies, suggests proprioception triggers alternation—a mechanism observed in rhythmic behaviors like locust flight and mammalian respiration¹¹⁰ [↗](#),¹¹¹ [↗](#),¹¹² [↗](#),¹¹³ [↗](#),¹¹⁴ [↗](#),¹¹⁵ [↗](#),¹¹⁶ [↗](#). Connections between position sensors, 13A neurons, and antagonistic MNs suggest that proprioceptive signals may trigger alternation of leg movements during grooming.

Local and descending neurons presynaptic to reciprocally connected 13A neurons could also induce alternation. The balance between internal circuits within the central nervous system and sensory feedback contributes to pattern generation. Future studies will dissect the extent of peripheral vs. central control in generating alternation.

These motifs can explain the spatial and temporal dynamics of grooming movements. **Flexors and extensors at multiple joints must coordinate to fully extend or contract a leg.** Neurons targeting related MNs could facilitate synchronization. For example, cluster 6 or 8 13A neurons target proximal and medial joint extensor MNs, allowing leg flexion. During head sweeps, proximal and medial joints move independently—for example, the proximal joint flexes while the medial joint extends— which can be coordinated by cluster 10 13A neurons and 13A-9e, -9f.

Flexors and extensors should be mutually exclusive

A generalist connected to both proximal and medial joint MNs could facilitate leg rubbing. For example, 13A-10c synchronously inhibits proximal, medial and distal flexor MNs (Figures 3C [↗](#) and Figure 2—figure supplement 1I4 [↗](#)) while targeting 13A neurons (13A-8a, -8b, and 8c) connected to extensor MNs (Figures 3F [↗](#) and Figure 2—figure supplement 1 G1-G3 [↗](#)). This arrangement ensures inhibition of MNs and disinhibition of antagonistic MNs across multiple joints, preventing simultaneous coactivation.

Flexors and extensors alternate or co-contract

Reciprocal inhibition and proprioceptive feedback onto 13A neurons could facilitate alternation between extension and flexion. Reciprocal inhibition between generalist neurons of 13As-i and 13As-ii could induce alternation during head sweeps (Figure 3D [↗](#)). This aligns with half center model which proposes that rhythmic motor patterns arise from two mutually inhibitory neuronal populations that alternate their activity in an out-of-phase manner to drive opposing muscle groups¹¹⁷ [↗](#),¹¹⁸ [↗](#),¹¹⁹ [↗](#). Our behavior experiments and modeling further support this connectivity, as we indeed induced rhythmic motion through pulsed activation of specific inhibitory neurons without altering any excitatory drive. However, flexor–extensor co-contraction can also be functionally relevant, such as for modulating joint stiffness during postural stabilization or for generating large forces required for fast movements¹²⁰ [↗](#),¹²¹ [↗](#),¹²² [↗](#). Some generalist 13A neurons could facilitate co-contraction across different leg segments, but none target antagonistic motor neurons controlling the same joint. Therefore, co-contraction within a single joint would require the simultaneous activation of multiple 13A neurons, potentially coordinated by upstream neurons that can recruit multiple 13As.

Inhibitory Innervation Imbalance Between Flexors and Extensors

We observed an imbalance in the innervation: more 13A neurons target extensors than flexors across multiple joints. While legs alternate between extension and flexion, they remain elevated during grooming. To maintain this posture, some MNs must be continuously activated while their antagonists are inactivated. Uneven distribution of inhibition could ensure that while some MNs remain active, others alternate flexion and extension in a controlled manner. Among the 13A neurons inhibiting antagonistic muscles, reciprocally connected ones could induce alternation, while others could keep legs elevated.

The asymmetry in connections, with more 13B neurons disinhibiting flexor-inhibiting 13As, suggests a mechanism for preferential flexion, supporting the flexor burst generator model, where the generator actively excites flexor MNs while inhibiting tonically active extensors^{123,124}.

It is important to note that our interpretations are based on connectome-derived connectivity patterns and behavioral observations from optogenetic manipulations, without direct physiological recordings from motor neurons or muscles. Studies in locust, stick insect, and cockroaches have demonstrated that fast cyclic leg movements can emerge from the interaction of rhythmic activity in a single muscle with the passive tension of its antagonist^{121,125,126,127,128}. Therefore, while our connectome analysis suggests motifs for multi-joint coordination, the exact patterns of motor neuron and muscle activity during grooming remain to be empirically confirmed.

Grooming Command Neuron Pathways to 13A Neurons

Descending neurons that command grooming movements, such as aDN¹²⁹ and DNg12¹¹, synapse onto 13A neurons. Although it remains unclear how constitutive activity in these descending neurons generates rhythmic grooming, the reciprocal inhibition circuits among 13As could provide a mechanism. The antennal grooming command neuron aDN synapses onto two primary 13As (γ and α ; 13As-i) that connect to proximal extensor and medial flexor motor neurons, as well as four other 13As (9a, 9c, 9i, 6e) projecting to body wall extensor motor neurons. These 13As-i also form reciprocal connections with 13As-ii, potentially supporting oscillatory leg movements. aDN connects to homologous 13As on both sides, consistent with the bilateral coordination required for antennal sweeping.

The head grooming/leg rubbing command neuron DNg12 synapses onto ~50 13As, primarily those projecting to proximal motor neurons.

While structural connectivity highlights candidate pathways for rhythmic movement generation, the dense interconnections among command neurons and premotor circuits suggest that multiple motifs likely contribute to the observed behaviors. Further work is needed to determine how these pathways are dynamically recruited during natural grooming sequences.

Spatial mapping of premotor neurons in the nerve cord

The organization of 13A neurons—where morphology, position, and motor-neuron connectivity align—suggests a premotor topographic map in the fly VNC. While 13A neurons partially reflect the myotopic map of MN dendrites^{18,19,130}, they do not target single MNs. Instead, they form clusters that connect either to a few MNs within a segment (specialists) or to groups spanning multiple leg segments and joints (generalists). Broadly projecting generalists could recruit or inhibit coordinated sets of muscles, effectively encoding multi-joint movement patterns, whereas specialists provide more precise, segment-specific control. In other words, this map represents both the leg-muscle architecture and premotor synergies—spatially organized modules for complex actions.

Such spatial logic is consistent with other mapped systems: sensory maps in the antennal lobe preserve receptor identity¹³¹,¹³²,¹³³, proprioceptors form an orderly representation of joint angles and forces in the VNC¹³⁴, or vertebrate spinal cords display dorsoventral recruitment gradients that scale locomotor output¹³⁵,¹³⁶. Together, these findings suggest that the fly VNC contains a layered spatial organization—from proprioceptive inputs to premotor and motor outputs—that could allow descending and sensory pathways to flexibly engage movement modules and synergies by addressing specific regions of neuropil. This raises the possibility that modular movement primitives encoded in the VNC can be combined to generate diverse leg behaviors, including grooming and walking.

Computational modeling of inhibitory circuits

We employ a linear rate-based (non-spiking) neural network to represent the 13A circuitry. Natural behavior and neural activity are of course more complex. Our model explores how 13A circuit motifs could contribute to, or produce *de novo*, behavioral features, including rhythmic movements (which might be otherwise produced by upstream circuits), in 2D space, matching the dimensionality of our behavioral data. This abstraction is inspired by Jovanic et al, 2016⁹³, but we add the agent component to the neural circuitry, and apply evolutionary “fine-tuning” of parameters. Similar circuit motifs, namely reciprocal inhibitions between pre-motor neurons and the sensory feedback have been modeled before, in particular neuroWalknet, where such motifs do not require a separate CPG component to generate rhythmic behavior⁴,⁹¹. However, our neuronal model is much simpler than the neuroWalknet - it controls a 2D agent operating on an abstract environment (the dust distribution). In real animals or complex mechanical models such as NeuroMechFly¹³⁷,¹³⁸, a more explicit central rhythm generation may be advantageous for the coordination across more degrees of freedom. When we perturb the model by activating 13A neurons with varying pulse lengths, we observe decreased coordination and increased movement frequency (compare angular velocities in **Figure 6J** with **Figure 6K**). Longer pulses can almost completely paralyze a leg (**Fig. 6K**). Here we did not attempt to simulate the exact experimental procedures. In future, the model’s parameters could be fine-tuned within similarly constrained space, but with the fitness function modified: instead of parameters being optimized to remove “dust”, they could be optimized by the similarity of behavioral features between the model and real flies, as has been done in whole-animal modeling¹³⁹, under various experimental conditions.

Future Directions

Our work lays groundwork for future exploration of the functional contributions of inhibitory circuits to motor control. Developing genetic tools to target specific inhibitory neurons and functional imaging during behavior would allow us to correlate temporal neuron activation with limb motion. While we focus on inhibitory control of one leg, manipulating these neurons reveals defects in left-right coordination. Investigating the circuitry involved—possibly mediated by commissural and/or descending neurons connected to these circuits—remains to be explored to elucidate the underlying circuits. Pulsed activation of 13A neurons induces grooming or walking in clean flies, while manipulating their activity in dusted flies alters grooming timing. This suggests that 13A neurons may be part of central pattern generators. While our experiments with multiple genetic lines labeling 13A/B neurons consistently implicate these cells in leg coordination, ectopic expression in some lines raises the possibility that other neurons may also contribute to these phenotypes. In addition, other excitatory and inhibitory neural circuits, not yet identified, may also contribute to the generation of rhythmic leg movements. Future studies should identify such neurons that regulate rhythmic timing and their interactions with inhibitory circuits.

Our connectome analysis reveal that inhibitory neurons within a hemilineage form circuit motifs with complex connections to specific leg motor neurons. These neurons likely complement excitatory premotor circuits, enabling multifunctionality in behaviors such as grooming and walking. Their connectivity suggests roles in coordinating movements across multiple joints,

enforcing flexor/extensor muscle antagonism, and driving extension and flexion alternation. We model and extrapolate potential functions of these complex inhibitory motifs. Normal activity of these inhibitory neurons is important for grooming; silencing or continuously activating them reduces time spent and effectiveness in dust removal, leading to forced flexion or abnormal extension, with limbs locked in extreme poses. Moreover, we show that inhibition, independent of excitatory input, plays an instructive role in generating rhythmic leg movements. Although we focus on grooming behavior, we expect these motifs to also contribute to walking, as the leg must recruit limited motor and premotor components to generate diverse movements. We conclude that inhibitory neurons are essential for controlling flexible limb movements and may play a key instructive role in timing and coordinating rhythmic behaviors.

Materials and methods

Contact for reagent and resource sharing

For information and inquiries regarding resources and reagents, please write to the lead contact Julie H. Simpson (jhsimpson@ucsb.edu).

Experimental model and subject details

Drosophila melanogaster were raised on a standard cornmeal medium at 25°C in a 12 hr light/dark cycle. For optogenetic experiments, one day old flies were transferred to food containing 0.4 mM all-trans-retinal and kept in the dark for 3 days. Genotypes of the fly lines are included in the Key Resources Table.

Identification of fly lines that target inhibitory neurons

We visually screened the VNC expression of various GAL4 lines on Flylight database¹⁴⁰ and compared them to the inhibitory hemilineages⁴¹. Next we obtained corresponding *AD*, and *DBD* flies from BDSC for the candidate lines and crossed them with *GAD-GAL4-AD* or *GAD-DBD* (Haluk Lacin) to confirm and restrict their expression in GABAergic neurons. *R35G04-GAL4-DBD*, *GAD-GAL4-AD* labeled six 13A and six 13B neurons per hemisegment. We also isolated six 13A neurons from this line by using *R35G04-GAL4-DBD*, *DBX-GAL4-AD* combination. *R11C07-GAL4-DBD* and *GAD-GAL4-AD* labels 4 inhibitory neurons. We intersected *R11C07 DBD* with *Dbx AD* to isolate two 13A neurons. *R11B07-GAL4-DBD*, *GAD-GAL4-AD* labeled 3 13B neurons.

Immunofluorescence and confocal microscopy

Flies were immobilized by anesthetizing them on ice (4°C). Central nervous system (CNS) was carefully dissected in 1X Phosphate-buffered saline (PBS). Subsequently, the wings were removed, and the flies were positioned ventral side up on a Sylgard plate. All legs were excised, and fine forceps (No. 5 Dumont from FST, Switzerland) were employed to delicately open the thorax along the midline, taking care to avoid damaging the underlying thoracic ganglia. A small incision near the first abdominal segment ensured preservation of the abdominal ganglion. Surrounding tissues were cleared from the thoracic ganglia, which were then gently extracted by grasping the neck connective. The dissected thoracic ganglia were subsequently fixed in 4% buffered paraformaldehyde for 45 minutes at 4°C.

Post-fixation, the thoracic ganglia underwent three 15-minute washes in 0.1% Triton X-100 (PBT) at room temperature on a shaker at 60 rpm, followed by a 20-minute wash in 0.1% PBT with normal goat serum (blocking solution). Primary antibodies, diluted in 0.1% PBT-NGS, were applied to the samples and incubated overnight at 4°C on a horizontal shaker. Following primary antibody incubation, the samples underwent three 15-minute washes in 0.1% PBT and one 20-minute wash in 0.1% PBT-NGS. Secondary antibodies, diluted in 0.1% PBT-NGS, were added to the samples and

incubated for 2-4 hours at room temperature on a shaker at 60 rpm. Secondary antibody removal was achieved through four 15-minute washes with 0.1% PBT at room temperature. Finally, the tissues were mounted on glass slides using Vectashield mounting medium (Vector Labs).

Primary antibodies used were Chicken pAb anti-GFP (Abcam, 1:1000), Rabbit (Rb) anti-GFP (Abcam, 1:1000), mouse (ms) anti-Neuroglian (BP104) (DSHB, 1:40), ms monoclonal anti-Brp antibody (nC82) (DSHB, 1:200). For MCFO labeling experiments, Rb mAb anti-HA (Cell Signaling Technologies, 1:300), Rat anti-FLAG (Novus Biologicals, 1:200), DyLight549-conjugated anti-V5 (AbD Serotec; 1:300 dilution).

Secondary antibodies from Invitrogen Molecular Probes conjugated with Alexa-488, Alexa-568 and Alexa-647 raised against chicken, ms and Rb were used in 1:400 dilution.

Zeiss LSM710 confocal microscope was used to obtain images of the CNS. Images were then processed in FIJI.

Recording and analysis of grooming in clean and dusted flies

For open field assay, we dusted the flies and obtained the recording as previously described⁴. Constant light intensity of 5.6 mW/cm² was used for continuous activation. Automated behavior analysis (ABRS) was used to quantify the amount of time flies perform individual grooming actions¹⁴¹. Additionally, manual scoring was performed² in flies showing uncoordinated leg movements. Quantification and statistical analysis describing the percentage of time dusted flies spent doing grooming and uncoordinated leg movements upon 13A and B activity manipulation was performed in Matlab as previously described⁹.

For limb tracking, either clean or dusted freely moving male flies were put in a studio containing 10-mm diameter quartz chamber and 100 Hz videos were recorded from below using FLIR Blackfly S camera. Custom-built LED panels (LXM2-PD01-0050, 625 nm) were utilized to deliver light activation from below, with an intensity of 1.1 mW/cm². Green led was used for silencing experiments.

Behavior analysis

Raw data, consisting of coordinates of the six annotated points on the front legs and two reference points on the body (per one frame of a video), was obtained from DeepLabCut⁷⁶. From these coordinates we computed: 1) (fly-centric) spatial positions of the body parts, 2) the spatial velocities of the points, 3) the whole-body velocity (the translation, obtained from the reference points in absolute coordinates), 4) the euclidean distances between the leg points (“joints”) and other “joints” or the reference points, and 5) the joint angles.

All behavior analysis was performed using Python, version 3.9.7.

Continuous feature extraction

The euclidean distances between various body parts (the six “joints” and the two reference points) are used as continuous features. The body velocity was computed as a euclidean distance of a point covered across a 50 frame (0.5 second) time window. Three joint angles per leg were computed from the three points on the leg and the two reference points. These angles are: posterior, medial, and distal angles. Angular velocities were computed as derivatives of the raw angles, and were lightly smoothed by a Gaussian filter (filter sigma = 2 frames). The euclidean distances were also smoothed by the same method.

The angular velocity time-series (AV) is used as the basic signal from which other continuous behavioral features are extracted, and also serves as the basis for segmentation. The main continuous features extracted from the AV are the phase and the frequencies.

The phase (between the movement of the front-legs) was computed from cross-correlation of AV signals of the two legs, using the `signal.correlation_lags()` function with the window for cross-correlation of five frames. We took the time-position of the maximum cross-correlation (the peak) as the lag (the phase). If the two legs are perfectly in-phase, the peak of the cross-correlation will be at time 0. The phase is a good indicator of a general type of grooming behaviors: front-leg rubbing is usually associated with a non-zero phase (a lag) whereas the head-cleaning more often than not has a zero lag (legs moving in-phase).

Frequencies are computed from the AV signal as well. We use `numpy.fft.fft()` function with the time window of 25 frames (0.25 seconds) and the input size of 160 frames (the length of the signal with the zero-padding) to compute the spectrum. The frequency at which the spectral power is maximal, is taken as the frequency of the signal.

Segmentation

Our analysis is focused on individual leg movements during grooming. These include contractions and extensions of the front-leg joints. For this purpose we separate the continuous features into segments corresponding to these movements.

The segmentation is performed by applying a stationary threshold to the angular velocity (AV) of the proximal joint angles. These joints determine the whole leg movements (flexions and extensions) so we consider them suitable for the purpose of segmentation. Next, for each segment we compute the averages, minima, and maxima of each continuous feature within that segment. We also compute the duration of each segment as an additional feature. These *segmented features* are used for behavioral classification and for further analysis, including comparing different groups of flies.

Classification of behavior and dimensionality reduction

We collect segments (segment features) from data-sets of groups of flies and pool them together for the purpose of classification. (e.g., experimental and control data-sets are used together.) Altogether we use 63 segment features, including averages of: euclidean distances, AVs, phases between joint pairs, frequencies of all joints, whole-body velocity, segment durations. We also include the maxima of the six AVs, and we can include combined features too (here we only use one combined feature – AV*segment duration).

For better mapping and classification of segments, we account for the temporal context of a segment: for each segment's set of features, we add the same type of features of the next two segments (in time). Thus we analyze features of triplets of segments rather than single segments.

The feature matrix X of the size $f \times t$, where t is the total number of segments and f is the number of segment features multiplied by 3 (to account for the temporal context), is the input to the UMAP dimensionality reduction algorithm.

We use UMAP from <https://pypi.org/project/umap-learn/> with the following parameters:

`n_neighbors=350,`

`min_dist=0.1,`

`n_components=2`

The **X** is thus projected onto two dimensions. For the UMAP construction we can remove some of the data from **X**. Namely, we can remove the data-points (segments) where angular velocity of proximal joints is below a certain value (fly is presumably not moving) or when the quality of data is too low (see quality control).

For the classification we apply the agglomerative clustering onto the 2D UMAP projection. We use `AgglomerativeClustering()` from `sklearn.cluster` (<https://scikit-learn.org/stable/modules/clustering.html#hierarchical-clustering>). We select the number of clusters of 12-14. Thus, we produce 12-14 class labels that can be applied to the 63 features of the segment feature matrix **X**.

Now we can compare the different behavioral classes, across the 63 features, between various groups of flies.

Comparing groups of flies

With multiple behavioral classes and features we can now compare different experimental groups of flies to assess the behavioral effects of the experimental procedures. For each *class-feature pair* we can determine whether there is a significant difference between the groups. Experimental and control groups contain data (segments) from multiple animals, so the segments from the same animal are not independent (clustered data). We therefore applied Linear Mixed Models (LMMs) to comparisons between the groups. This was accomplished by the Python method `mixedlm()` from `statsmodels.formula.api` library (<https://www.statsmodels.org/stable/api.html>).

The groups were compared across several features-class pairs. This calls for the multiple hypotheses adjustment. We applied the False Discovery Rate method, using the `statsmodels.stats.multitest` function (<https://www.statsmodels.org/dev/generated/statsmodels.stats.multitest.multipletests.html>). The alpha parameter was set to 0.05, and the method argument was set to `fdr_bh`.

Model

The 13A circuits

The model of 13A and associated circuits was build from simple linear neuronal networks. We did not model individual neurons but rather abstracted them to nodes, interconnected by synaptic weights corresponding to the numbers of synapses obtained from the connectome analysis. Since, the 13A circuits in the right and left front leg neuropil are mirrored and there are slightly more connections on the left side (*Figure 2—figure supplement 2*), the synaptic weights of the same circuits on the left neuropil were used for modeling purposes.

The network is shown in **Figure 6A**. Below is the description of the 13A circuit model for one leg. The full model consists of two (front) legs, build the same way.

The two 13A nodes connect to each other reciprocally (inhibitory synapses). The adjacency matrix for these nodes is:

$$W_{13A \leftrightarrow 13A} = \begin{bmatrix} 0 & -559 \\ -488 & 0 \end{bmatrix}$$

The activation levels of the 13As are simply:

$$A_{13A}^{t+1} = A_{13A}^t + W_{13A \leftrightarrow 13A}^T * A_{13A}^t - \theta_{13A}$$

A_{13A} is a vector of activation levels of the 13As, and **θ**_{13A} is a vector of thresholds of 13A neurons.

The 13A nodes inhibit the six motor neurons (MNs). These connections are represented by adjacency matrix:

$$W_{13A \rightarrow MN} = \begin{bmatrix} -1594.0 & -1525.0 & -1525.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & -1525.0 & -1594.0 & -1594.0 \end{bmatrix}$$

The six MNs act on the three pairs of “antagonistic muscles”, which in turn control the changes of joint angles (three per leg). Flexion and extension rates of the antagonistic muscles are directly proportional to the activation levels of the MNs.

The activation levels of the MNs are:

$$A_{MN}^{t+1} = A_{MN}^t + W_{13A \rightarrow MN}^T * A_{13A}^t - \Theta_{MN}$$

A_{MN} is a vector of activation levels of the MNs, A_{13A} is a vector of activation levels of 13As, and Θ_{MN} is a vector of thresholds of 13A neurons.

The inputs to the six sensory neurons (SNs) are directly proportional to the flexion and extension rates of the joint muscles. So, the activation levels of the SNs are:

$$A_{SN}^{t+1} = A_{SN}^t + extension/flexionrate - \Theta_{SN},$$

where A_{SN} is a vector of activation levels of SNs, and Θ_{SN} is a vector of thresholds of SNs.

The SNs then feed back onto both, the MNs and the 13As. The adjacency matrix representing the sensory feedback to the MNs is:

$$W_{SN \rightarrow MN} = \begin{bmatrix} 0.0 & 0.0 & 0.0 & 244.0 & 244.0 & 244.0 \\ 0.0 & 0.0 & 0.0 & 244.0 & 244.0 & 244.0 \\ 0.0 & 0.0 & 0.0 & 244.0 & 244.0 & 244.0 \\ 548.0 & 548.0 & 548.0 & 0.0 & 0.0 & 0.0 \\ 548.0 & 548.0 & 548.0 & 0.0 & 0.0 & 0.0 \\ 548.0 & 548.0 & 548.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}$$

The activation levels of the MNs are then updated as follows:

$$A_{MN}^{t+1} = A_{MN}^t + W_{SN \rightarrow MN}^T * A_{SN}^t - \Theta_{MN}$$

where A_{MN} is a vector of activation levels of MNs, and Θ_{MN} is a vector of thresholds of MNs.

The adjacency matrix representing the sensory feedback to the 13As is:

$$W_{SN \rightarrow 13A} = \begin{bmatrix} 526.0 & 0.0 \\ 526.0 & 0.0 \\ 526.0 & 0.0 \\ 153.0 & 48.0 \\ 153.0 & 48.0 \\ 153.0 & 48.0 \end{bmatrix}$$

And the A_{13A} activation levels are updated:

$$A_{13A}^{t+1} = A_{13A}^t + W_{SN \rightarrow 13A}^T * A_{SN}^t$$

Activation levels are constrained as follows:

$$- \max(A_{13A}) = 2500$$

$$- \max(A_{MN}; A_{SN}) = 200$$

Minimal activation levels for all neurons are set at zero (so, no negative rates).

Two 13B nodes were added to inhibit the two 13A nodes. The initial weights of the 13B → 13A were -413 and -160.

The excitatory network

The model network also needs an excitatory input. Because we do not know the upstream excitatory connections from the connectome we created a “black box” network that takes the simulated dust distribution as the input, and it outputs excitatory signal to the 13A, 13B, and MNs. Other inputs include: the “proprioceptive” inputs (“efference copy” from motor neurons), and the amount of dust accumulated on both legs. The “black box” excitatory network consists of a recurrent neural network (RNN) as the hidden layer, with linear synapses and initially random weights.

The hidden RNN layer has 40 nodes. The input layer is the dust grid and the pixel values are its inputs. The output layer has 21 nodes connecting to the two 13B nodes, the two 12A nodes, and the 6 MN nodes, *per one leg* (so 10 outputs per leg). The same excitatory network also feeds to the other leg’s 13A network, in the same manner. The remaining output node does not project anywhere (it is placed there for future model development where it could output the amplitude of exploratory noise injected into the 13A network).

The agent and the environment

Our model is composed of neuronal circuits embedded in a simple agent that acts on its environment. The agent has two 2D legs, corresponding to the two front legs of a fly, each of which is made of three points (“joints”): proximal, medial, and distal. The distal point (the one farthest from the “body”) can remove the “dust” from the environment. The movement of the “joints” (per leg) is controlled by three pairs of “antagonistic muscles” affecting the three angles formed by the “joints.” **Figure 6I** [↗](#) shows a frame of a movie where the legs are represented by the three points each.

The environment is composed of a Gaussian distribution of “dust” around the agent (green pixels in **Figure 6I** [↗](#)). The means of the Gaussian are at the center of the 32 by 32 pixel grid (also the position of the “root” of the front legs), so $x=0$; $y=0$. The variance = 5 pixels, in both directions. The maximal amount of dust (at the peak of the Gaussian) is 1.0 (0.99).

The agent can remove the “dust” when the distal “joint” sweeps over the environment with the minimum velocity of 1 pixel/frame. (So, if the “joint” just stays at a given position, the dust is not getting removed. It has to move over it.)

As the grooming behavior is being performed, the “dust” accumulates on the legs, reducing their ability to continue removing it from the grid (the “body”). The “leg cleaning” - removal of the accumulated dust occurs when the two legs are in proximity to each other (Euclidean distance < 5 pixels). The “leg cleaning” rate of dust removal from legs is the same as the body dust removal rate: 0.5/frame.

The three angles (per leg) are constrained. Distal and medial angles: 100° - 180°; proximal angle: 80° - 120°.

The fine-tuning of the synaptic weights

When we run the model with the default synaptic weights (see previous sections) nothing happens, i.e. the activation levels either saturate (reach the ceiling values shown above) fall to zero, or reach a steady value. The legs of the agent may move once or twice and then the model “freezes.” One way of getting around this problem would be to add a “CPG component” to the model, to drive the periodic excitatory inputs, thereby creating a baseline periodic activity and movements. We could then study the effects of the 13A circuitry on these movements. However, we wanted to see if the model network could generate periodic movements all by itself.

The empirical weights used in the model are approximate and we do not know the exact ratios between individual modeled weights (the assumption is that the number of biological synapses corresponds to the weights). So, we allow the modeled weights to vary (in value, but not in sign) and yet preserve the approximate ratios obtained from the empirical data. In other words, we are exploring a space of possible models that adhere to approximately the weight ratios obtained empirically. Specifically, the weights are allowed to vary $\pm 20\%$ and cannot change the sign – i.e. inhibitory neurons must remain inhibitory. (In future work we may decrease this space of exploration to adhere even more closely to the empirical estimates. Conversely, we may increase the exploration space and observe all solutions to see how close our empirical estimates are to the global optimum.)

We thus constrain the model to the connectome as follows:

The connectome matrix reference $\mathbf{W}_{\text{ref}}$ contains the empirical weights for a given neuron type (e.g. $\mathbf{W}_{13A \rightarrow 13A}$) and $\mathbf{W}_{\text{mod}}$ contains the weights from the evolving model (see below). First we eliminate non-zero weights from the model ($\mathbf{W}_{\text{mod}}$) that should be zero in the reference matrix $\mathbf{W}_{\text{ref}}$:

$$M_{ij} = \begin{cases} 1 & \text{if } W_{\text{ref},ij} \neq 0 \\ 0 & \text{if } W_{\text{ref},ij} = 0 \end{cases}$$

$$\mathbf{W}_{\text{masked}} = M \odot \mathbf{W}_{\text{raw}}$$

Because the absolute numbers in the empirical matrices ($\mathbf{W}_{\text{ref}}$) are meaningless for the modeling we can rescale them -assuming that same the ratios of the synaptic weights between the same types of neurons are preserved. So, each adjacency matrix connecting the same type of neurons is scaled separately. For example, $\mathbf{W}_{13A \rightarrow \text{MN}}$ is specifying the connections between 13As and MNs, and is thus scaled separately from $\mathbf{W}_{13A \rightarrow 13A}$ which specifies connections between 13As. Each $\mathbf{W}_{\text{ref}}$ is scaled as shown:

$$a_{\text{raw}} = \max_{i,j} |\mathbf{W}_{\text{masked},ij}|, \quad a_{\text{ref}} = \max_{i,j} |\mathbf{W}_{\text{ref},ij} \cdot M_{ij}|$$

$$\mathbf{W}_{\text{ref}}^{\text{scaled}} = \frac{a_{\text{raw}}}{a_{\text{ref}}} \mathbf{W}_{\text{ref}}$$

We then compute the tolerance bounds, where tolerance $\tau = 0.2$:

$$\mathbf{W}_{\text{min}} = (1 - \tau) \mathbf{W}_{\text{ref}}^{\text{scaled}}, \quad \mathbf{W}_{\text{max}} = (1 + \tau) \mathbf{W}_{\text{ref}}^{\text{scaled}}$$

Finally we enforce sign constraints and clip the model matrix $\mathbf{W}_{\text{mod}}$:

$$\mathbf{W}_{\text{signed},ij} = |\mathbf{W}_{\text{masked},ij}| \text{sgn}(\mathbf{W}_{\text{ref},ij})$$

$$\mathbf{W}_{\text{mod},ij} = \begin{cases} \min \left(\max(\mathbf{W}_{\text{signed},ij}, \mathbf{W}_{\text{min},ij}), \mathbf{W}_{\text{max},ij} \right) & \text{if } M_{ij} = 1 \\ 0 & \text{if } M_{ij} = 0 \end{cases}$$

To explore the space around our empirically estimated weights, we employed a genetic algorithm (GA) that we wrote (see the model code at <https://github.com/PrimoRavbar/Inhibitory-circuits>). Briefly, we evolved a population of 250-500 genomes containing the model parameters (weight matrices, thresholds) and hyper-parameters (max firing rates). The mutation rate was set to 0.1. The genomes were competing in cohorts of 6. At each epoch (each genome – agent – in the cohort played 2 “games”) fitness rates were assigned and the strongest genomes were allowed to reproduce accordingly. *There was no cross-over*. Importantly, for every new generation the mutated genomes were adjusted to fit the empirical weights within the tolerance limit of 0.2 (clipped, as described above). This way the evolutionary exploration remained bounded throughout the process.

Connectome analysis

Neuronal reconstruction, lineage identification and detection of neuronal partners

We used serial-section transmission electron microscopy (TEM) dataset of female adult *Drosophila* (FANC)²⁰ to reconstruct 13A and 13B hemilineages in the VNC. These neurons were identified in the EM volume based on their cell body clusters, arborization pattern and nerve bundle entry positions into the ventral neuropil^{40,41,81} and comparison with light-level images and axonal tracts labeled with anti-Neuroglian.

13A neurons cluster together and enter the VNC neuropil anteriorly through the ventrolateral position^{40,41,81,142}. 13B neurons have contralateral cell bodies and ipsilateral projections, with their axons entering the neuropil through the extreme ventral bundle^{40,41,81}. Using confocal microscopy images of 13A and 13B neurons marked with GFP and axonal tracts labeled with anti-Neuroglian for reference comparison, we located these neurons in the EM volume. We manually traced the main neural skeletons and later proof-read automatic segmentations. Then neuronal IDs and cell body coordinates of each 13A and 13B neurons is shown in Table S1.

Manual reconstructions of some of the 13A and 13B neurons were initially performed in CATMAID¹⁴³. Traced skeletons were then imported from CATMAID to Neuroglancer¹⁴⁴. We identified other 13A neurons and 13B neurons in the corresponding hemilineage bundles and proofread errors in the automated neuronal reconstructions¹⁶. We fully proofread 62 13A neurons (Table S1), 64 13B neurons in the right prothoracic segment (T1) of VNC, and 25/64 13B neurons in the left T1. We used the automated synapse detection to identify the downstream and upstream connections¹⁶. We used various FANC packages¹⁶ generously available to the community to generate upstream and downstream partner summary of all the 13A and 13B neurons in R studio (https://rdrr.io/github/flyconnectome/fanc/man/fanc_partner_summary.html).

Connectivity matrix

To plot connectivity matrices between groups of neurons, we utilized Python libraries including pandas, networkx, and matplotlib. We created a directed graph using networkx to represent the connections, where presynaptic and postsynaptic neurons were added as nodes. The thickness and color of edges between nodes were determined by the strength of the connections, and the type of presynaptic or postsynaptic neurons, respectively. Node colors were assigned based on the type of neurons, with specific colors denoting different subtypes of 13A/B neurons and MNs. Finally, we generated the visualization using matplotlib. 13B to 13A connections were manually added in **Figure 3B**. Leg schematic and MN to muscle connections were also manually added in Adobe Illustrator.

Cosine similarity matrix

We computed the cosine similarity matrix of 13A neurons based on their downstream motor connections in Python using the `cosine_similarity` from `sklearn.metrics.pairwise` for computing cosine similarities. A pivot table was created from the DataFrame, with neurons as rows (index) and their post-synaptic targets (`post_id`) as columns. The values in this table represented the weights of the connections. Duplicates were aggregated using the `sum` function, and missing values were filled with zeros. The cosine similarity between each pair of neurons was calculated using the `cosine_similarity` function. Cosine similarity is a measure that calculates the cosine of the angle between two vectors. In this context, each neuron is represented as a vector of its connectivity weights to downstream MNs. The cosine similarity value ranges from -1 to 1, where: 1 indicates that the vectors are identical. 0 indicates that the vectors are orthogonal (no similarity). -1 indicates that the vectors are diametrically opposed. This calculation resulted in a similarity matrix, where each entry (*i*, *j*) represents the cosine similarity between the connectivity profiles of neuron *i* and neuron *j*. This calculation resulted in a similarity matrix, where each entry (*i*, *j*) represents the cosine similarity between the connectivity profiles of neuron *i* and neuron *j*. The resulting cosine similarity matrix was visualized using `matplotlib`. The matrix was displayed as a heatmap with a color gradient indicating the degree of similarity.

Classification of 13A and 13B neurons based on morphology

We used NBLAST^{16,86}, a computational method to measure pairwise similarity between neurons based on their position and geometry to identify various subclasses within the hemilineages. We performed hierarchical clustering on pairwise NBLAST similarity scores computed using `navis.nblast_allbyall()`. The resulting similarity matrix was symmetrized by averaging it with its transpose, and converted into a distance matrix using the transformation:

$$\text{distance} = (1 - \text{similarity}) \quad \text{distance} = (1 - \text{similarity})$$

This ensures that a perfect NBLAST match (similarity = 1) corresponds to a distance of 0.

Clustering was performed using Ward's linkage method (`method='ward'` in `scipy.cluster.hierarchy.linkage`), which minimizes the total within-cluster variance and is well-suited for identifying compact, morphologically coherent clusters. We did not predefine the number of clusters. Instead, clusters were visualized using a dendrogram, where branch coloring is based on the default behavior of `scipy.cluster.hierarchy.dendrogram()`. By default, this function applies a visual color threshold at 70% of the maximum linkage distance to highlight groups of similar elements. In our dataset, this corresponded to a linkage distance of approximately 1–1.5, which visually separated morphologically distinct neuron types (**Figures 2A** and **Figure 2—figure supplement 3A**). This threshold was used only as a visual aid and not as a hard cutoff for quantitative grouping. This classification was based on similarity scores and included left-right comparisons.

Data availability

Data is available in supplementary figures and code is available on GitHub.

Resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Chicken polyclonal anti-GFP	Abcam	RRID: AB_300798
Rabbit (Rb) polyclonal anti- GFP	Invitrogen	Cat #A11122; RRID: AB_221569
Mouse (ms) monoclonal anti-Bruchpilot	DSHB	RRID: AB_2314866
ms anti-Neuroglian (BP104)	DSHB	RRID: AB_528402
Mouse polyclonal anti-V5:DyLight 550	AbD Serotec	RRID: AB_2687576
Rabbit polyclonal anti-HA	Cell Signaling Technologies	RRID: AB_1549585
Rat monoclonal anti-FLAG	Novus Bio	RRID: AB_1625982
Goat anti-Chicken Alexa Fluor 488	Invitrogen	RRID:AB_142924
Goat anti-rabbit Alexa Fluor 488	Invitrogen	RRID. AB_143165
Goat anti-mouse Alexa Fluor 568	Invitrogen	RRID: AB_2534072

Goat anti-mouse Alexa Fluor 633	Invitrogen	RRID:AB_2535719
Goat anti-rabbit Alexa Fluor 568	Invitrogen	RRID: AB_143157
Goat anti-rat Alexa Fluor 488	Invitrogen	RRID: AB_2534074
Donkey anti-rat Alexa 647	Jackson ImmunoResearch	RRID: AB_2340694
Chemicals, Peptides, and Recombinant Proteins		
Insect-a-slip	Bio Quip Products	Cat#2871A
Reactive Yellow 86	Organic Dyestuffs Corporation	CAS 61951-86-8
Experimental Models: Organisms/Strains		
R35G04-GAL4-DBD	Bloomington Stock Center (BDSC)	RRID: BDSC_70351
GAD-GAL4-AD	Deng et al., 2019	
GAD-GAL4-DBD	Gift from Haluk Lacin and James Truman	

Resources table (continued)

Dbx-GAL4-AD	Gift from Haluk Lacin and James Truman	
Dbx-GAL4-DBD	Gift from Haluk Lacin and James Truman	
R11C07 AD	BDSC	RRID: BDSC_70533
w[1118]; P{y[+t7.7] w[+mC]=20XUAS-IVS- CsChrimson.mVenus} attP40	BDSC	RRID: BDSC_55135
P{JFRC7-20XUAS-IVS- mCD8::GFP} attp40	BDSC	RRID: BDSC_32194
UAS-GTACR1	Gift from Adam Claridge-Chang	
w[*]; P{y[+t7.7] w[+mC]=UAS- GtACR1.d.EYFP}attP2	BDSC	RRID: BDSC_92983

Resources table (continued)

w[1118] P{y[+t7.7] w[+mC]=R57C10-FLPL} su(Hw)attP8; PBac{y[+mDint2] w[+mC]=10xUAS(FRT.sto p)myr::smGdP-HA} VK00005 P{y[+t7.7] w[+mC]=10xUAS(FRT.sto p)myr::smGdP-V5- THS-10xUAS(FRT.stop)m yr::smGdP-FLAG} su(Hw)attP1(MCFO3)	BDSC	RRID: BDSC_64087
10XUAS-IVS-eGFP-Kir2.1	von Reyn et al., 2014	
Control-GAL4-AD-GAL4- DBD EMPTY SPLIT: BPp65ADZp(attp40); BPZpGDBD(attp2)	BDSC	RRID: BDSC_79603
Software and Algorithms		
DeepLabCut	Mathis et al., 2018	RRID:SCR_021391
Python		RRID:SCR_008394
MATLAB	MathWorks	RRID: SCR_001622
FIJI	Schindelin et al., 2012	RRID:SCR_002285
Adobe illustrator		RRID:SCR_010279

Resources table (continued)

Adobe Photoshop		RRID:SCR_014199
Braincircuits		https://braincircuits.io/app?p=fruitfly_fanc_public
Neuroglancer	Maitin-Shepard et al., 2021	RRID:SCR_015631
fancr	FANC packages	https://github.com/flyconnectome/fancr
neuPrint	Plaza et al., 2022	https://neuprint.janelia.org/
CATMAID	Saalfeld et al., 2009	RRID:SCR_006278
RStudio		RRID:SCR_000432

Resources table (continued)

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Additional information

Author contributions

D.S.S.: Conceptualization, Investigation, Methodology, Formal analysis, Writing – original draft, review and editing. P.R.: Modeling, Methodology, Software, Formal analysis. J.H.S.: Conceptualization, Funding acquisition, Supervision, Writing – review and editing.

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Additional files

Supplementary 

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Reviewer #1 (Public review):

Summary:

Syed et al. investigate the circuit underpinnings for leg grooming in the fruit fly. They identify two populations of local interneurons in the right front leg neuromere of ventral nerve cord, i.e. 62 13A neurons and 64 13B neurons. Hierarchical clustering analysis identifies each 10 morphological classes for both populations. Connectome analysis reveals their circuit interactions: these GABAergic interneurons provide synaptic inhibition either between the two subpopulations, i.e. 13B onto 13A, or among each other, i.e. 13As onto other 13As, and/or onto leg motoneurons, i.e. 13As and 13Bs onto leg motoneurons. Interestingly, 13A interneurons fall into two categories with one providing inhibition onto a broad group of motoneurons, being called "generalists", while others project to few motoneurons only, being called "specialists". Optogenetic activation and silencing of both subsets strongly effects leg grooming. As well activating or silencing subpopulations, i.e. 3 to 6 elements of the 13A and 13B groups has marked effects on leg grooming, including frequency and joint positions and even interrupting leg grooming. The authors present a computational model with the four circuit motifs found, i.e. feed-forward inhibition, disinhibition, reciprocal inhibition and redundant inhibition. This model can reproduce relevant aspects of the grooming behavior.

Strengths:

The authors succeeded in providing evidence for neural circuits interacting by means of synaptic inhibition to play an important role in the generation of a fast rhythmic insect motor behavior, i.e. grooming. Two populations of local interneurons in the fruit fly VNC comprise four inhibitory circuit motifs of neural action and interaction: feed-forward inhibition, disinhibition, reciprocal inhibition and redundant inhibition. Connectome analysis identifies the similarities and differences between individual members of the two interneuron populations. Modulating the activity of small subsets of these interneuron populations markedly affects generation of the motor behavior thereby exemplifying their important role for generating grooming. The authors carefully discuss strengths and limitations of their approaches and place their findings into the broader context of motor control.

Weaknesses:

Effects of modulating activity in the interneuron populations by means of optogenetics were conducted in the so-called closed-loop condition. This does not allow to differentiate between direct and secondary effects of the experimental modification in neural activity, as feedforward and feedback effects cannot be disentangled. To do so open loop experiments, e.g. in deafferented conditions, would be important. Given that many members of the two populations of interneurons do not show one, but two or more circuit motifs, it remains to be disentangled which role the individual circuit motif plays in the generation of the motor behavior in intact animals.

Comments on revisions:

The careful revision of the manuscript improved the clarity of presentation substantially.

<https://doi.org/10.7554/eLife.106446.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript by Syed et al. presents a detailed investigation of inhibitory interneurons, specifically from the 13A and 13B hemilineages, which contribute to the generation of rhythmic leg movements underlying grooming behavior in *Drosophila*. After performing a detailed connectomic analysis, which offers novel insights into the organization of premotor inhibitory circuits, the authors build on this anatomical framework by performing optogenetic perturbation experiments to functionally test predictions derived from the connectome. Finally, they integrate these findings into a computational model that links anatomical connectivity with behavior, offering a systems-level view of how inhibitory circuits may contribute to grooming pattern generation.

Strengths:

- (1) Performing an extensive and detailed connectomic analysis, which offers novel insights into the organization of premotor inhibitory circuits.
- (2) Making sense of the largely uncharacterized 13A/13B nerve cord circuitry by combining connectomics and optogenetics is very impressive and will lay the foundation for future experiments in this field.
- (3) Testing the predictions from experiments using a simplified and elegant model.

Weaknesses:

- (1) In Figure 4-figure supplement 1, the inclusion of walking assays in dusted flies is problematic, as these flies are already strongly biased toward grooming behavior and rarely walk. To assess how 13A neuron activation influences walking, such experiments should be conducted in undusted flies under baseline locomotor conditions.
- (2) Regarding Fig 5: The 70ms on/off stimulation with a slow opsin seems problematic. CsChrimson off kinetics are slow and unlikely to cause actual activity changes in the desired neurons with the temporal precision the authors are suggesting they get. Regardless, it is amazing the authors get the behavior! It would still be important for authors to mention the optogenetics caveat, and potentially supplement the data with stimulation at different frequencies, or using faster opsins like ChrimsonR.

Overall, I think the strengths outweigh the weaknesses, and I consider this a timely and comprehensive addition to the field.

Reviewer #3 (Public review):

Summary:

The authors set out to determine how GABAergic inhibitory premotor circuits contribute to the rhythmic alternation of leg flexion and extension during *Drosophila* grooming. To do this, they first mapped the ~120 13A and 13B hemilineage inhibitory neurons in the prothoracic segment of the VNC and clustered them by morphology and synaptic partners. They then tested the contribution of these cells to flexion and extension using optogenetic activation and inhibition and kinematic analyses of limb joints. Finally, they produced a computational model representing an abstract version of the circuit to determine how the connectivity identified in EM might relate to functional output. The study makes important contributions to the literature.

The authors have identified an interesting question and use a strong set of complementary tools to address it:

They analysed serial-section TEM data to obtain reconstructions of every 13A and 13B neuron in the prothoracic segment. They manually proofread over 60 13A neurons and 64 13B neurons, then used automated synapse detection to build detailed connectivity maps and cluster neurons into functional motifs.

They used optogenetic tools with a range of genetic driver lines in freely behaving flies to test the contribution of subsets of 13A and 13B neurons.

They used a connectome-constrained computational model to determine how the mapped connectivity relates to the rhythmic output of the behavior.

<https://doi.org/10.7554/eLife.106446.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

Syed et al. investigate the circuit underpinnings for leg grooming in the fruit fly. They identify two populations of local interneurons in the right front leg neuromere of ventral nerve cord, i.e. 62 13A neurons and 64 13B neurons. Hierarchical clustering analysis identifies 10 morphological classes for both populations. Connectome analysis reveals their circuit interactions: these GABAergic interneurons provide synaptic inhibition either between the two subpopulations, i.e., 13B onto 13A, or among each other, i.e., 13As onto other 13As, and/or onto leg motoneurons, i.e., 13As and 13Bs onto leg motoneurons. Interestingly, 13A interneurons fall into two categories, with one providing inhibition onto a broad group of motoneurons, being called "generalists", while others project to a few motoneurons only, being called "specialists". Optogenetic activation and silencing of both subsets strongly affect leg grooming. As well as activating or silencing subpopulations, i.e., 3 to 6 elements of the 13A and 13B groups, has marked effects on leg grooming, including frequency and joint positions, and even interrupting leg grooming. The authors present a computational model with the four circuit motifs found,

i.e., feed-forward inhibition, disinhibition, reciprocal inhibition, and redundant inhibition. This model can reproduce relevant aspects of the grooming behavior.

Strengths:

The authors succeeded in providing evidence for neural circuits interacting by means of synaptic inhibition to play an important role in the generation of a fast rhythmic insect motor behavior, i.e., grooming. Two populations of local interneurons in the fruit fly VNC comprise four inhibitory circuit motifs of neural action and interaction: feed-forward inhibition, disinhibition, reciprocal inhibition, and redundant inhibition. Connectome analysis identifies the similarities and differences between individual members of the two interneuron populations. Modulating the activity of small subsets of these interneuron populations markedly affects the generation of the motor behavior, thereby exemplifying their important role in generating grooming.

We thank the reviewer for their thoughtful and constructive evaluation of our work.

Weaknesses:

Effects of modulating activity in the interneuron populations by means of optogenetics were conducted in the so-called closed-loop condition. This does not allow for differentiation between direct and secondary effects of the experimental modification in neural activity, as feedforward and feedback effects cannot be disentangled. To do so, open loop experiments, e.g., in deafferented conditions, would be important. Given that many members of the two populations of interneurons do not show one, but two or more circuit motifs, it remains to be disentangled which role the individual circuit motif plays in the generation of the motor behavior in intact animals.

Our optogenetic experiments show a role for 13A/B neurons in grooming leg movements – in an intact sensorimotor system - but we cannot yet differentiate between central and reafferent contributions. Activation of 13As or 13Bs disinhibits motor neurons and that is sufficient to induce walking/grooming. Therefore, we can show a role for the disinhibition motif.

Proprioceptive feedback from leg movements could certainly affect the function of these reciprocal inhibition circuits. Given the synapses we observe between leg proprioceptors and 13A neurons, we think this is likely.

Our previous work (Ravbar et al 2021) showed that grooming rhythms in dusted flies persist when sensory feedback is reduced, indicating that central control is possible. In those experiments, we used dust to stimulate grooming and optogenetic manipulation to broadly silence sensory feedback. We cannot do the same here because we do not yet have reagents to separately activate sparse subsets of inhibitory neurons while silencing specific proprioceptive neurons. More importantly, globally silencing proprioceptors would produce pleiotropic effects and severely impair baseline coordination, making it difficult to distinguish whether observed changes reflect disrupted rhythm generation or secondary consequences of impaired sensory input. Therefore, the reviewer is correct – we do not know whether the effects we observe are feedforward (central), feedback sensory, or both. We have included this in the revised results and discussion section to describe these possibilities and the limits of our current findings.

Additionally, we have used a computational model to test the role of each motif separately and we show that in the results.

Reviewer #2 (Public review):

Summary:

This manuscript by Syed et al. presents a detailed investigation of inhibitory interneurons, specifically from the 13A and 13B hemilineages, which contribute to the generation of rhythmic leg movements underlying grooming behavior in Drosophila. After performing a detailed connectomic analysis, which offers novel insights into the organization of premotor inhibitory circuits, the authors build on this anatomical framework by performing optogenetic perturbation experiments to functionally test predictions derived from the connectome. Finally, they integrate these findings into a computational model that links anatomical connectivity with behavior, offering a systems-level view of how inhibitory circuits may contribute to grooming pattern generation.

Strengths:

- (1) Performing an extensive and detailed connectomic analysis, which offers novel insights into the organization of premotor inhibitory circuits.*
- (2) Making sense of the largely uncharacterized 13A/13B nerve cord circuitry by combining connectomics and optogenetics is very impressive and will lay the foundation for future experiments in this field.*
- (3) Testing the predictions from experiments using a simplified and elegant model.*

We thank the reviewer for their thoughtful and encouraging evaluation of our work.

Weaknesses:

- (1) In Figure 4, while the authors report statistically significant shifts in both proximal inter-leg distance and movement frequency across conditions, the distributions largely overlap, and only in Panel K (13B silencing) is there a noticeable deviation from the expected 7-8 Hz grooming frequency. Could the authors clarify whether these changes truly reflect disruption of the grooming rhythm?*

We reanalyzed the dataset with Linear Mixed Models. We find significant differences in mean frequencies upon silencing these neurons but not upon activation. The experimental groups are also significantly more variable. We revised these panels with updated analysis. We think these data do support our interpretation that the grooming rhythms are disrupted.

- More importantly, all this data would make the most sense if it were performed in undusted flies (with controls) as is done in the next figure.*

In our assay conditions, undusted flies groom infrequently. We used undusted flies for some optogenetic activation experiments, where the neuron activation triggers behavior initiation, but we chose to analyze the effect of silencing inhibitory neurons in dusted flies because dust reliably activates mechanosensory neurons and elicits robust grooming behavior enabling us to assess how manipulation of 13A/B neurons alters grooming rhythmicity and leg coordination.

- (2) In Figure 4-Figure Supplement 1, the inclusion of walking assays in dusted flies is problematic, as these flies are already strongly biased toward grooming behavior and rarely walk. To assess how 13A neuron activation influences walking, such experiments should be conducted in undusted flies under baseline locomotor conditions.*

We agree that there are better ways to assay potential contributions of 13A/13B neurons to walking. We intended to focus on how normal activity in these inhibitory neurons affects coordination during grooming, and we included walking because we observed it in our optogenetic experiments and because it also involves rhythmic leg movements. The walking data is reported in a supplementary figure because we think this merits further study with assays designed to quantify walking specifically. We will make these goals clearer in the revised manuscript and we are happy to share our reagents with other research groups more equipped to analyze walking differences.

(3) For broader lines targeting six or more 13A neurons, the authors provide specific predictions about expected behavioral effects-e.g., that activation should bias the limb toward flexion and silencing should bias toward extension based on connectivity to motor neurons. Yet, when using the more restricted line labeling only two 13A neurons (Figure 4 - Figure Supplement 2), no such prediction is made. The authors report disrupted grooming but do not specify whether the disruption is expected to bias the movement toward flexion or extension, nor do they discuss the muscle target. This is a missed opportunity to apply the same level of mechanistic reasoning that was used for broader manipulations.

Because we cannot unambiguously identify one of the neurons from our sparsest 13A splitGAL4 lines in FANC, we cannot say with certainty which motor neurons they target. That limits the accuracy of any functional predictions.

(4) Regarding Figure 5: The 70ms on/off stimulation with a slow opsin seems problematic. CsChrimson off kinetics are slow and unlikely to cause actual activity changes in the desired neurons with the temporal precision the authors are suggesting they get. Regardless, it is amazing that the authors get the behavior! It would still be important for the authors to mention the optogenetics caveat, and potentially supplement the data with stimulation at different frequencies, or using faster opsins like ChrimsonR.

We were also intrigued by the behavioral consequences of activating these inhibitory neurons with CsChrimson. We appreciate the reviewer's point that CsChrimson's slow off-kinetics limit precise temporal control. To address this, we repeated our frequency analysis using a range of pulse durations (10/10, 50/50, 70/70, 110/110, and 120/120 ms on/off) and compared the mean frequency of proximal joint extension/flexion cycles across conditions. We found no significant difference in frequency (LLMS, $p > 0.05$), suggesting that the observed grooming rhythm is not dictated by pulse period but instead reflects an intrinsic property of the premotor circuit once activated. We now include these results in 'Figure 5—figure supplement 1' and clarify in the text that we interpret pulsed activation as triggering, rather than precisely pacing, the endogenous grooming rhythm. We continue to note in the manuscript that CsChrimson's slow off-kinetics may limit temporal precision. We will try ChrimsonR in future experiments.

Overall, I think the strengths outweigh the weaknesses, and I consider this a timely and comprehensive addition to the field.

Reviewer #3 (Public review):

Summary:

The authors set out to determine how GABAergic inhibitory premotor circuits contribute to the rhythmic alternation of leg flexion and extension during Drosophila grooming. To do this, they first mapped the ~120 13A and 13B hemilineage inhibitory neurons in the

prothoracic segment of the VNC and clustered them by morphology and synaptic partners. They then tested the contribution of these cells to flexion and extension using optogenetic activation and inhibition and kinematic analyses of limb joints. Finally, they produced a computational model representing an abstract version of the circuit to determine how the connectivity identified in EM might relate to functional output. The study, in its current form, makes an important but overclaimed contribution to the literature due to a mismatch between the claims in the paper and the data presented.

Strengths:

The authors have identified an interesting question and use a strong set of complementary tools to address it:

(1) They analysed serial-section TEM data to obtain reconstructions of every 13A and 13B neuron in the prothoracic segment. They manually proofread over 60 13A neurons and 64 13B neurons, then used automated synapse detection to build detailed connectivity maps and cluster neurons into functional motifs.

(2) They used optogenetic tools with a range of genetic driver lines in freely behaving flies to test the contribution of subsets of 13A and 13B neurons.

(3) They used a connectome-constrained computational model to determine how the mapped connectivity relates to the rhythmic output of the behavior.

Weaknesses:

The manuscript aims to reveal an instructive, rhythm-generating role for premotor inhibition in coordinating the multi-joint leg synergies underlying grooming. It makes a valuable contribution, but currently, the main claims in the paper are not well-supported by the presented evidence.

Major points

(1) Starting with the title of this manuscript, "Inhibitory circuits generate rhythms for leg movements during *Drosophila* grooming", the authors raise the expectation that they will show that the 13A and 13B hemilineages produce rhythmic output that underlies grooming. This manuscript does not show that. For instance, to test how they drive the rhythmic leg movements that underlie grooming requires the authors to test whether these neurons produce the rhythmic output underlying behavior in the absence of rhythmic input. Because the optogenetic pulses used for stimulation were rhythmic, the authors cannot make this point, and the modelling uses a "black box" excitatory network, the output of which might be rhythmic (this is not shown). Therefore, the evidence (behavioral entrainment; perturbation effects; computational model) is all indirect, meaning that the paper's claim that "inhibitory circuits generate rhythms" rests on inferred sufficiency. A direct recording (e.g., calcium imaging or patch-clamp) from 13A/13B during grooming - outside the scope of the study - would be needed to show intrinsic rhythmogenesis. The conclusions drawn from the data should therefore be tempered. Moreover, the "black box" needs to be opened. What output does it produce? How exactly is it connected to the 13A-13B circuit?

We modified the title to better reflect our strongest conclusions: "Inhibitory circuits control leg movements during *Drosophila* grooming"

Our optogenetic activation was delivered in a patterned (70 ms on/off) fashion that entrains rhythmic movements, but this does not rule out the possibility that the rhythm is imposed externally. In the manuscript, we state that we used pulsed light to mimic a flexion-extension cycle and note that this approach tests whether inhibition is sufficient to drive rhythmic leg

movements when temporally patterned. While this does not prove that 13A/13B neurons are intrinsic rhythm generators, it does demonstrate that activating subsets of inhibitory neurons is sufficient to elicit alternating leg movements resembling natural grooming and walking.

Our goal with the model was to demonstrate that it is possible to produce rhythmic outputs with this 13A/B circuit, based on the connectome. The “black box” is a small recurrent neural network (RNN) consisting of 40 neurons in its hidden layer. The inputs are the “dust” levels from the environment (the green pixels in Figure 6I), the “proprioceptive” inputs (“efference copy” from motor neurons), and the amount of dust accumulated on both legs. The outputs (all positive) connect to the 13A neurons, the 13B neurons, and to the motor neurons. We refer to it as the “black box” because we make no claims about the actual excitatory inputs to these circuits. Its function is to provide input, needed to run the network, that reflects the distribution of “dust” in the environment as well as the information about the position of the legs.

The output of the “black box” component of the model might be rhythmic. In fact, in most instances of the model implementation this is indeed the case. However, as mentioned in the current version of the manuscript: “But the 13A circuitry can still produce rhythmic behavior even without those external inputs (or when set to a constant value), although the legs become less coordinated.” Indeed, when we refine the model (with the evolutionary training) without the “black box” (using a constant input of 0.1) the behavior is still rhythmic and sustained. Therefore, the rhythmic activity and behavior can emerge from the premotor circuitry itself without a rhythmic input.

The context in which the 13A and 13B hemilineages sit also needs to be explained. What do we know about the other inputs to the motorneurons studied? What excitatory circuits are there?

We agree that there are many more excitatory and inhibitory, direct and indirect, connections to motor neurons that will also affect leg movements for grooming and walking. 13A neurons provide a substantial fraction of premotor input. For example, 13As account for ~17.1% of upstream synapses for one tibia extensor (femur seti) motor neuron and ~14.6% for another tibia extensor (femur feti) motor neuron. Our goal was to demonstrate what is possible from a constrained circuit of inhibitory neurons that we mapped in detail, and we hope to add additional components to better replicate the biological circuit as behavioral and biomechanical data is obtained by us and others.

Furthermore, the introduction ignores many decades of work in other species on the role of inhibitory cell types in motor systems. There is some mention of this in the discussion, but even previous work in Drosophila larvae is not mentioned, nor crustacean STG, nor any other cell types previously studied. This manuscript makes a valuable contribution, but it is not the first to study inhibition in motor systems, and this should be made clear to the reader.

We thank the reviewer for this important reminder. Previous work on the contribution of inhibitory neurons to invertebrate motor control certainly influenced our research. We have expanded coverage of the relevant history and context in our revised discussion.

(2) The experimental evidence is not always presented convincingly, at times lacking data, quantification, explanation, appropriate rationales, or sufficient interpretation.

We are committed to improving the clarity, rationale, and completeness of our experimental descriptions. We have revisited the statistical tests applied throughout the manuscript and expanded the Methods.

(3) The statistics used are unlike any I remember having seen, essentially one big t-test followed by correction for multiple comparisons. I wonder whether this approach is optimal for these nested, high-dimensional behavioral data. For instance, the authors do not report any formal test of normality. This might be an issue given the often skewed distributions of kinematic variables that are reported. Moreover, each fly contributes many video segments, and each segment results in multiple measurements. By treating every segment as an independent observation, the non-independence of measurements within the same animal is ignored. I think a linear mixed-effects model (LMM) or generalized linear mixed model (GLMM) might be more appropriate.

We thank the reviewer for raising this important point regarding the statistical treatment of our segmented behavioral data. Our initial analysis used independent t-tests with Bonferroni correction across behavioral classes and features, which allowed us to identify broad effects. However, we acknowledge that this approach does not account for the nested structure of the data. To address this, we re-analyzed key comparisons using linear mixed-effects models (LMMs) as suggested by the reviewer. This approach allowed us to more appropriately model within-fly variability and test the robustness of our conclusions. We have updated the manuscript based on the outcomes of these analyses.

(4) The manuscript mentions that legs are used for walking as well as grooming. While this is welcome, the authors then do not discuss the implications of this in sufficient detail. For instance, how should we interpret that pulsed stimulation of a subset of 13A neurons produces grooming and walking behaviours? How does neural control of grooming interact with that of walking?

We do not know how the inhibitory neurons we investigated will affect walking or how circuits for control of grooming and walking might compete. We speculate that overlapping pre-motor circuits may participate because both have similar extension flexion cycles at similar frequencies, but we do not have hard experimental data to support. This would be an interesting area for future research. Here, we focused on the consequences of activating specific 13A/B neurons during grooming because they were identified through a behavioral screen for grooming disruptions, and we had developed high-resolution assays and familiarity with the normal movements in this behavior.

(5) The manuscript needs to be proofread and edited as there are inconsistencies in labelling in figures, phrasing errors, missing citations of figures in the text, or citations that are not in the correct order, and referencing errors (examples: 81 and 83 are identical; 94 is missing in text).

We have proofread the manuscript to fix figure labeling, citation order, and referencing errors.

Reviewing Editor Comments:

In addition to the recommendations listed below, a common suggestion, given the lack of evidence to support that 13A and 13B are rhythm-generating, is to tone down the title to something like, for example, "Inhibitory circuits control leg movements during grooming in Drosophila" (or similar).

We changed the title to Inhibitory circuits control leg movements during Drosophila grooming

Reviewer #1 (Recommendations for the authors):

(1) Naming of movements of leg segments:

The authors refer to movements of leg segments across the leg, i.e., of all joints, as "flexion" and "extension". For example, in Figure 4A and at many other places. This naming is functionally misleading for two reasons: (i) the anatomical organization of an insect leg differs in principle from the organization of the mammalian leg, which the manuscript often refers to. While the organization of a mammalian limb is planar the organization of the insect limb shows a different plane as compared to the body length axis (for detailed accounts see Ritzmann et al. 2004; Büschges & Ache, 2024); (ii) the reader cannot differentiate between places in the text, where "flexion" and "extension" refer to movements of the tibia of the femur-tibia joint, e.g. in the graphical abstract, in Figure 3 and its supplements, and other places, e.g. Figure 4 and its supplements, where these two words refer to movements of leg segments of other joints, e.g. thorax-coxa, coxa-trochanter and tarsal joints. The reviewer strongly suggests naming the movements of the leg segments according to the individual joint and its muscles.

We accept this helpful suggestion. We now include a description of the leg segments and joints in the revised Introduction and refer to which leg segments we mean

“The adult *Drosophila* leg consists of serially arranged joints—bodywall/thoraco-coxal (Th-C), coxa–trochanter (C-Tr), trochanter–femur (Tr-F), femur–tibia (F-Ti), tibia–tarsus (Ti-Ta)—each powered by opposing flexor and extensor muscles that transmit force through tendons (Soler et al., 2004). The proximal joints, Th-C and C-Tr, mediate leg protraction–retraction and elevation–depression, respectively (Ritzmann et al., 2004; Büschges & Ache, 2025). The medial joint, F-Ti, acts as the principal flexion–extension hinge and is controlled by large tibia extensor motor neurons and flexor motor neurons (Soler et al., 2004; Baek and Mann 2009; Brierley et al., 2012; Azevedo et al., 2024; Lesser et al., 2024). By contrast, distal joints such as Ti-Ta and the tarsomeres contribute to fine adjustments, grasping, and substrate attachment (Azevedo et al., 2024).”

We also clarified femur-tibia joints in the graphical abstract, modified Figure 3 legend and added joints at relevant places.

(2) Figures 3, 4, and 5 with supplements:

The authors optogenetically silence and activate (sub)populations of 13A and 13B interneurons. Changes in frequency of movements and distance between legs or leg movements are interpreted as the effect of these experimental paradigms. No physiological recordings from leg motoneurons or leg muscles are shown. While I understand the notion of the authors to interpret a movement as the outcome of activity in a muscle, it needs to be remembered that it is well known that fast cyclic leg movements, including those for grooming, cannot be used to conclude on the underlying neural activity. Zakotnik et al. (2006) and others provided evidence that such fast cyclic movements can result from the interaction of the rhythmic activity of one leg muscle only, together with the resting tension of its silent antagonist. Given that no physiological recordings are presented, this needs to be mentioned in the discussion, e.g., in the section "Inhibitory Innervation Imbalance.....".

Added studies from Heitler, 1974; Bennet-Clark, 1975; Zakotnik et al., 2006; Page et al., 2008 in discussion.

(3) Introduction and Discussion:

The authors refer extensively to work on the mammalian spinal cord and compare their own work with circuit elements found in the spinal cord. From the perspective of the reviewer this notion is in conflict with acknowledging prior research work on the role of inhibitory network interactions for other invertebrates and lower vertebrates: such are locust flight system (for feedforward inhibition, disinhibition), crustacean stomatogastric nervous system (reciprocal inhibition), clone swimming system (reciprocal inhibition, feedforward inhibition, disinhibition), leech swimming system (reciprocal inhibition, disinhibition, feedforward inhibition), xenopus swimming system (reciprocal inhibition). The next paragraph illustrates this criticism/suggestion for stick insect neural circuits for leg stepping.

(4) Discussion:

"Feedforward inhibition" and "Disinhibition": it is already been described that rhythmic activity of antagonistic insect leg motoneuron pools arises from alternating synaptic inhibition and disinhibition of the motoneurons from premotor central pattern generating networks, e.g., Büschges (1998); Büschges et al. (2004); Ruthe et al. (2024).

We have added these references to the revised Discussion.

(5) Circuit motifs of the simulation, i.e., mutual inhibition between interneurons and onto motoneurons and sensory feedback influences and pathways share similarities to those formerly used by studies simulating rhythmic insect leg movements, for example, Schilling & Cruse 2020, 2023 or Toth et al. 2012. For the reader, it appears relevant that the progress of the new simulation is explained in the light of similarities and differences to these former approaches with respect to the common circuit motifs used.

We now put our work in the context of other models in the Discussion section: “Similar circuit motifs, namely reciprocal inhibitions between pre-motor neurons and the sensory feedback have been modeled before, in particular neuroWalknet, and such simple motifs do not require a separate CPG component to generate rhythmic behavior in these models (Schilling & Cruse 2020, 2023). However, our model is much simpler than the neuroWalknet - it controls a 2D agent operating on an abstract environment (the dust distribution), without physics. In real animals or complex mechanical models such as NeuroMechFly (Lobato-Rios et al), a more explicit central rhythm generation may be advantageous for the coordination across many more degrees of freedom.”

Reviewer #2 (Recommendations for the authors):

I might have missed this, but I couldn't find any mention of how the grooming command pathways, described by previous work from the authors' lab, recruit these predicted grooming pattern-generating neurons. This should be mentioned in the connectome analysis and also discussed later in the discussion.

13A neurons are direct downstream targets of previously described grooming command neurons. Specifically, the antennal grooming command neuron aDN (Hampel et al., 2015) synapses onto two primary 13As (γ and α ; 13As-i) that connect to proximal extensor and medial flexor motor neurons, as well as four other 13As (9a, 9c, 9i, 6e) projecting to body wall extensor motor neurons. The 13As-i also form reciprocal connections with 13As-ii, providing a potential substrate for oscillatory leg movements. aDN connects to homologous 13As on both sides, consistent with the bilateral coordination needed for antennal sweeping.

The head grooming/leg rubbing command neuron DNg12 (Guo et al., 2022) synapses directly onto ~50 13As, predominantly those connected to proximal motor neurons.

While sometimes the structural connectivity suggests pathways for generating rhythmic movements, the extensive interconnections among command neurons and premotor circuits indicate that multiple motifs could contribute to the observed behaviors. Further work will be needed to determine how these inputs are dynamically engaged during normal grooming sequences. We have now added it to the discussion.

I encourage the authors to be explicit about caveats wherever possible: e.g., ectopic expression in genetic tools, potential for other unexplored neurons as rhythm generators (rather than 13A/B), given that the authors never get complete silencing phenotypes, CsChrimson kinetics, neurotransmitter predictions, etc.

We now explain these caveats as follows: Ectopic expression is noted in Figure 1—figure supplement 1, and we added the following to the Discussion: “While our experiments with multiple genetic lines labeling 13A/B neurons consistently implicate these cells in leg coordination, ectopic expression in some lines raises the possibility that other neurons may also contribute to this phenotype. In addition, other excitatory and inhibitory neural circuits, not yet identified, may also contribute to the generation of rhythmic leg movements. Future studies should identify such neurons that regulate rhythmic timing and their interactions with inhibitory circuits.”

We also added a caveat regarding CsChrimson kinetics in the Results. Finally, our identification of these neurons as inhibitory is based on genetic access to the GABAergic population (we use GAD-spGAL4 as part of the intersection which targets them), rather than on predictions of neurotransmitter identity.

Reviewer #3 (Recommendations for the authors):

Detailed list of figure alterations:

(1) Figure 1:

(a) Figure 1B and Figure 1 - Figure Supplement 1 lack information on individual cells - how can we tell that the cells targeted are indeed 13A and 13B, and which ones they are? Since off-target expression in neighboring hemilineages isn't ruled out, the interpretation of results is not straightforward.

The neurons labeled by R35G04-DBD and GAD1-AD are identified as 13A and 13B based on their stereotyped cell body positions and characteristic neurite projections into the neuropil, which match those of 13A and 13B neurons reconstructed in the FANC and MANC connectome. While we have not generated flip-out clones in this genotype, we do isolate 13A neurons more specifically later in the manuscript using R35G04-DBD intersected with Dbx-AD, and show single-cell morphology consistent with identified 13A neurons. The purpose of including this early figure was to motivate the study by showing that silencing this population, which includes 13A/13B neurons, strongly reduces grooming in dusted flies.

Regarding Figure 1—Figure Supplement 1:

This figure showed the expression patterns of all lines used throughout the manuscript. Panels C and D illustrated lines with minimal to no ectopic expression. Panels A and B show neurons with posterior cell bodies that may correspond to 13A neurons not reconstructed in our dataset but described in Soffers et al., 2025 and Marin et al., 2025 and we have provided detailed information about all VNC expressions in the figure legend.

(b) Figure 1D lacks explanation of boxplots, asterisks, genotypes/experimental design.

Added.

(c) Figures 1E-F and video 1 lack quantification, scale bars.

Added quantification.

(2) Figure 2:

(a) Figure 2A, Figure 2 - Supplement 3: What are the details of the hierarchical clustering? What metric was used to decide on the number of clusters?

We have used FANC packages to perform NBLAST clustering (Azevedo et al., 2024, Nature). We now include the full protocol in Methods. The details are as follows:

We performed hierarchical clustering on pairwise NBLAST similarity scores computed using `navis.nblast_allbyall()`. The resulting similarity matrix was symmetrized by averaging it with its transpose, and converted into a distance matrix using the transformation:

$$\text{distance} = (1 - \text{similarity}) \quad \text{distance} = (1 - \text{similarity})$$

This ensures that a perfect NBLAST match (similarity = 1) corresponds to a distance of 0.

Clustering was performed using Ward's linkage method (`method='ward'` in `scipy.cluster.hierarchy.linkage()`), which minimizes the total within-cluster variance and is well-suited for identifying compact, morphologically coherent clusters.

We did not predefine the number of clusters. Instead, clusters were visualized using a dendrogram, where branch coloring is based on the default behavior of `scipy.cluster.hierarchy.dendrogram()`. By default, this function applies a visual color threshold at 70% of the maximum linkage distance to highlight groups of similar elements. In our dataset, this corresponded to a linkage distance of approximately 1–1.5, which visually separated morphologically distinct neuron types (Figures 2A and Figure 2—figure supplement 3A). This threshold was used only as a visual aid and not as a hard cutoff for quantitative grouping.

The Methods section says that the classification "included left-right comparisons". What does that mean? What are the implications of the authors only having proofread a subset of neurons in T1L (see below)?

All adult leg motor neurons and 13A neurons (except one, 13A-ε) have neurite arbors restricted to the local, ipsilateral neuropil associated with the nearest leg. Although 13B neurons have contralateral cell bodies, their projections are also entirely ipsilateral. The Tuthill Lab, with contributions from our group, focused proofreading efforts on the left front neuropil (T1L) in FANC. This is also where the motor neuron to muscle mapping has been most extensively done. We reconstructed/proofread the 13A and 13B neurons from the right side as well (T1R). We see similar clustering based on morphology and connectivity here as well.

Reconstructions lack scale bars and information on orientation (also in other figures), and the figures for the 13B analysis are not consistent with the main figure (e.g., labelling of clusters in panel B along x,y axes).

Added.

(b) Figure 2B: Since the cosine similarity matrix's values should go from -1 to 1, why was a color map used ranging from 0 to 1?

While cosine similarity values can theoretically range from -1 to 1, in our case, all vector entries (i.e., synaptic weights) are non-negative, as they reflect the number of synapses from each 13A neuron to its downstream targets. This means all pairwise cosine similarities fall within the 0 to 1 range.

Why are some neurons not included in this figure, like 1g, 2b, 3c-f (also in Supplement 3)?

The few 13A neurons that don't connect to motor neurons are not shown in the figure.

(c) Figures 2C and D: the overlaid neurites are difficult to distinguish from one another. If the point here is to show that each 13A neuron class innervates specific motor neurons, then this is not the clearest way of doing that. For instance, the legend indicates that extensors are labelled in red, and that MNs with the highest number of synapses are highlighted in red - does that work? I could not figure out what was going on. On a more general point: if two cells are connected, does that not automatically mean that they should overlap in their projection patterns?

We intended these panels to illustrate that 13A neurons synapse onto overlapping regions of motor neurons, thereby creating a spatial representation of muscle targets. However, we agree that overlapping multiple neurons in a single flat projection makes the figure difficult to interpret. We have therefore removed Figures 2C and 2D.

While neurons must overlap at least somewhere if they form a synaptic connection, the amount of their neurites that overlap can vary, and more extensive overlap suggests more possible connections. Because the synapses are computationally predicted, examining the overlap helps to confirm that these predictions are consistent.

While connected neurons must overlap locally at their synaptic sites, they do not necessarily show extensive or spatially structured overlap of their projections. For example, descending neurons or 13B interneurons may form synapses onto motor neurons without exhibiting a topographically organized projection pattern. In contrast, 13A → MN connectivity is organized in a structured manner: specialist 13A neurons align with the myotopic map of MN dendrites, whereas generalist 13As project more broadly and target MN groups across multiple leg segments, reflecting premotor synergies. This spatial organization—combining both joint-specific and multi-joint representations—was a key finding we wished to highlight, and we have revised the Results text to make this clearer.

(d) Figure 2 - Figure Supplement 1: Why are these results presented in a way that goes against the morphological clustering results, but without explanation? Clusters 1-3 seem to overlap in their connectivity, and are presented in a mixed order. Why is this ignored? Are there similar data for 13B?

The morphological clusters 1–3 do exhibit overlapping connectivity, but this is consistent with both their anatomical similarity and premotor connectivity. Specifically, Cluster 1 neurons connect to SE and TrE motor neurons, Cluster 2 connects only to TrE motor neurons, and Cluster 3 targets multiple motor pools, including SE and TrE (Figure 2—Figure Supplement 1B). This overlap is also reflected in the high pairwise cosine similarity among Clusters 1–3 shown in Figure 2B. Thus, their similar connectivity profiles align with their proximity in the NBLAST dendrogram.

Regarding 13B neurons: there is no clear correlation between morphological clusters and downstream motor targets, as shown in the cosine similarity matrix (Figure 2—figure supplement 3). Moreover, even premotor 13B neurons that fall within the same morphological cluster do not connect to the same set of motor neurons (Figure 3—figure supplement 1F). For example, 13B-2a connects to LTrM and tergo-trochanteral MNs, 13B-2b connects to TiF MNs, and 13B-2g connects to Tr-F, TiE, and tergo-T MNs. Together, these results demonstrate that 13A neurons are spatially organized in a manner that correlates with their motor neuron targets, whereas 13B neurons lack such spatially structured organization, suggesting distinct principles of connectivity for these two inhibitory premotor populations.

(e) Figure 2 - Figure Supplement 2: A comparison is made here between T1R (proofread) and T1L (largely not proofread). A general point is made here that there are "similar numbers of neurons and cluster divisions". First, no quantitative comparison is provided, making it difficult to judge whether this point is accurate. Second, glancing at the connectivity diagram, I can identify a large number of discrepancies. How should we interpret those? Can T1L be proofread? If this is too much of a burden, results should be presented with that as a clear caveat.

The 13A and 13B neurons in the T1L hemisegment are fully proofread (Lesser et al, 2024, current publication); the T1R has been extensively analyzed as well. To compare the clustering and match identities of 13A and 13B neurons on the left and the right, We mirrored the 13A neurons from the left side and used NBLAST to match them with their counterparts on the right.

While individual synaptic counts differ between sides in the FANC dataset (T1L generally showing higher counts), the number of 13A neurons, their clustering, and the overall patterns of connectivity are largely conserved between T1L and T1R.

Importantly, each 13A cluster targets the same subset of motor neurons on both sides, preserving the overall pattern of connectivity. The largest divergence is seen in cluster 9, which shows more variable connectivity.

(f) Figure 2 - Figure Supplements 4 & 5: Why did the authors choose to present the particular cell type in Supplement 4? Why are the cell types in Supplement 5 presented differently? Labels in Supplement 5 are illegible, but I imagine this is due to the format of the file presented to reviewers. Why are there no data for 13B?

We chose to present the particular cell type in Supplement 4 because it corresponds to cell types targeted in the genetic lines used in our behavioral experiments. The 13A neuron shown is also one of the primary neurons in this lineage. This example illustrates its broader connectivity beyond the inhibitory and motor connections emphasized in the main figures.

In Supplement 5, we initially aimed to highlight that the major downstream targets of 13A neurons are motor neurons. We have now removed this figure and instead state in the text that the major downstream targets are MNs.

We did not present 13B neurons in the same format because their major downstream targets are not motor neurons. Instead, we emphasize their role in disinhibition and their connections to 13A neurons, as shown in a specific example in Figure 3—figure supplement 2. This 13B neuron also corresponds to a cell type targeted in the genetic line used in our behavioral experiments.

(3) Figure 3:

(a) Figure 3A: the collection of diagrams is not clear. I'd suggest one diagram with all connections included repeated for each subpanel, with each subpanel highlighting relevant connections and greying out irrelevant ones to the type of connection discussed. The nomenclature should be consistent between the figure and the legend (e.g., feedforward inhibition vs direct MN inhibition in A1.

The intent of Figure 3A is to highlight individual circuit motifs by isolating them in separate panels. Including all connections in every sub panel would likely reduce clarity and make it harder to follow each motif. For completeness, we show the full set of connections together in Panel D. We updated the nomenclature as suggested.

(b) Figure 3B: Why was the medial joint discussed in detail? Do the thicknesses of the lines represent the number of synapses? There should be a legend, in that case. Why are the green edges all the same thickness? Are they indeed all connected with a similarly low number of synapses?

We focused on the medial joint (femur-tibia joint) because it produces alternating flexion and extension of the tibia during both head sweeps and leg rubbing, which are the main grooming actions we analyzed. During head grooming, the tarsus is typically suspended in the air, so the cleaning action is primarily driven by tibial movements generated at the medial joint.

The thickness of the edges represents the number of synapses, and we have now clarified this in the legend. The green edges represent connections from 13B neurons, which were manually added to the graph, as described in the Methods section. 13B neurons are smaller than 13A neurons and form significantly fewer total downstream synapses. For example, the 13B neuron shown in Figure 3—figure supplement 2 makes a total of 155 synapses to all downstream neurons, with only 22 synapses to its most strongly connected partner, a 13A neuron. The relatively sparse connectivity of 13B neurons is shown in thinner or uniform edge weights in this graph.

(C) Figure 3C: This is a potentially important panel, but the connections are difficult to interpret. Moreover, the text says, "This organizational motif applies to multiple joints within a leg as reciprocal connections between generalist 13A neurons suggest a role in coordinating multi-joint movements in synergy". To what extent is this a representative result? The figure also has an error in the legend (it is not labelled as 3C).

This statement is true and based on the connectivity of these neurons. We now added

"Data for 13A-MN connections shown in Figure 2—figure supplement 1 I9, I6, I7, H9, H4, and H5; 13A-13A connections shown in Figure 3—figure supplement 1C." to the figure legend.

Thanks, we fixed the labelling error.

(d) Figure 3 - Figure Supplement 1: Panel A is very difficult to interpret. Could a hierarchical diagram be used, or some other representation that is easier to digest?

Panel A provides a consolidated view of all upstream and downstream interconnections among individual 13A and 13B neurons, allowing readers to quickly assess which neurons connect to which others without having to examine all subpanels. For a hierarchical representation, we have provided individual neuron-level diagrams in Panels C–F.

(e) Figure 3 - Figure Supplement 2: Why was this cell type selected?

We selected this 13B because it is involved in the disinhibition of 13A neurons and is also present in the genetic line used for our behavioral experiments.

(f) Figure 3 - Figure Supplement 3: The diagram is confusing, with text aligned randomly, and colors lacking some explanations. Legend has odd formatting.

The diagram layout and text alignment are designed to reflect the logical grouping of proprioceptors, 13A neurons, and motor neurons. To improve clarity, we have added node colors, included a written explanation for edge colors, and corrected the formatting of the figure legend.

(4) Figure 4:

(a) Figure 4A: This has no quantification, poor labelling, and odd units (centiseconds?). The colours between the left and right panels also don't align.

We have fixed these issues.

(b) Figure 4D-K: The ranges on the different axes are not the same (e.g., y axis on box plots, x axis on histograms). This obscures the fact that the differences between experimental and control, which in many cases are not big, are not consistent between the various controls. Moreover, the data that are plotted are, as far as I can tell (which is also to say: this should be explained), one value per frame. With imaging at 100Hz, this means that an enormous number of values are used in each analysis. Very small differences can therefore be significant in a statistical sense. However, how different something is between conditions is important (effect size), and this is not taken into account in this manuscript. For instance, in 4D-J, the differences in the mean seem to be minimal. Should that not be taken into consideration? A point in case is panel D in Figure 4 - Figure Supplement 1: even with near identical distributions, a statistically significant difference is detected. The same applies to Figure 4 - Figure Supplements 1-3. Also, what do the boxes and whiskers in the box plots show, exactly?

We have re-plotted all summary panels using linear mixed-effects models (LMMs) as suggested. In the updated plots, each dot represents the mean value for a single animal, and bar height represents the group mean. Whiskers indicate the 95% confidence interval around the group mean. This approach avoids inflating sample size by using per-frame values and provides a more accurate view of both variability and effect size.

(e) Figure 4 - Figure Supplement 1: There are 6 cells labelled in the split line; only 4 are shown in A3. Is cluster 6 a convincing match between EM and MCFO?

We indeed report four neurons targeted by the split-GAL4 line in flip out clones. Generating these clones was technically challenging. In our sample (n=23), we may not have labeled all of the neurons. Alternatively, two neurons may share very similar morphology and connectivity, making it difficult to tell them apart. We have added this clarification to the revised figure legend.

It is interesting to see data on walking in panel K, but why were these analyses not done on any of the other manipulations? What defect produced the reduction in velocity, exactly? How should this be interpreted?

Our primary focus was on grooming, but we did observe changes in walking, so we report illustrative examples. We initially included a panel showing increased walking velocity upon 13A activation, but this effect did not survive FDR correction and was removed in the revised

version. We instead included data for 13A silencing which did not affect the frequency of joint movements during walking. However, spatial aspects of walking were affected: the distance between front leg tips during stance was reduced, indicating that although flies continued to walk rhythmically, the positioning of the legs was altered. This suggests that these specific 13A neurons may influence coordination and limb placement during walking without disrupting basic rhythmicity. As reviewer #2 also noted, dust may itself affect walking, so we have chosen not to further pursue this aspect in the current study.

(f) Figure 4 - Figure Supplement 2: panel A is identical to Figure 1 - Figure Supplement 1C. This figure needs particular attention, both in content and style. Why present data on silencing these neurons in C-D, but not in E-F?

We removed the panel Figure 1 - Figure Supplement 1C and kept it in Figure 4 - Figure Supplement 2 A. E-F also shows data on silencing, as C'.

(g) Figure 4 - Figure Supplement 3: In panel B, the authors should more clearly demonstrate the identity of 4b and 4a. Why present such a limited number of parameters in F and G?

The cells shown in panel B represent the best matches we could identify between the light-level expression pattern and EM reconstructions. In panels F and G, we focused on bout duration, as leg position/inter-leg distance and frequency were already presented (in Figure 4). Together, these parameters demonstrate the role of 13B neurons in coordinating leg movements. Maximum angular velocity of proximal joints was not significantly affected and is therefore not included.

(5) Figure 5:

(a) Figure 5B: Lacks a quantification of the periodic nature of the behavior, which is required to compare to experimental conditions, e.g., in panel C.

Added

(b) Figure 5C: Requires a quantification; stimulus dynamics need to be incorporated.

Added

(c) Figure 5D: More information is needed. Does "Front leg" mean "leg rub", and "Head" "head sweep"? How do the dynamics in these behaviors compare to normal grooming behavior?

Yes, head grooming is head sweeps and Front leg grooming is leg rub. Comparison added, shown in 5E-F

(d) Figure 5E: How should we interpret these plots? Do these look like normal grooming/walking?

We have now included the comparison.

(e) Figure 5F: Needs stats to compare it to 5B'.

Done

(6) Figure 6:

(a) Figure 6A: I think the circuit used for the model is lacking the claw/hook extension - 13Bs connection. Any other changes? What is the rationale?

13Bs upstream of these particular 13As do not receive significant connections from claw/hook neurons (there's only one ~5 synapses connection from one hook extension to one 13B neurons, which we neglected for the modeling purpose).

(b) Figure 6B and C: Needs labels, legend; where is 13B?

In the figure legend we now added: "The 13B neurons in this model do not connect to each other, receive excitatory input from the black box, and only project to the 13As (inhibitory). Their weight matrix, with only two values, is not shown." We added the colorbar and corrected the color scheme.

(c) Figure 6D-H: plots are very difficult to interpret. Units are also missing (is "Time" correct?).

The units are indeed Time in frames (of simulation). We added this to the figure and the legend. We clarified the units of all variables in these panels. Corrected the color scheme and added their meaning to the legend text.

(d) Figure 6I: I think the authors should consider presenting this in a different format.

(e) Figure 6 J and K (also Figure Supplement): lacks labels.

We added labels for the three joints, increased the size of fonts for clarity, and added panel titles on the top.

More specific suggestions:

(1) It would be helpful if the titles of all figures reflected the take-away message, like in Figure 2.

(2) "Their dendrites occupy a limited region of VNC, suggesting common pre-synaptic inputs" - all dendrites do, so I'd suggest rephrasing to be more precise.

(3) "We propose that the broadly projecting primary neurons are generalists, likely born earlier, while specialists are mostly later-born secondary neurons" - this needs to be explained.

We added the explanation.

We propose that the broadly projecting primary neurons are generalists, likely born earlier, while specialists are mostly later-born secondary neurons. This is consistent with the known developmental sequence of hemilineages, where early-born primary neurons typically acquire larger arbors and integrate across broader premotor and motor targets, whereas later-born secondary neurons often have more spatially restricted projections and specialized roles [18,19,81,82,85]. Our morphological clustering supports this idea: generalist 13As have extensive axonal arbors spanning multiple leg segments, whereas specialist neurons are more narrowly tuned, connecting to a few MN targets within a segment. Thus, both their morphology and connectivity patterns align with the expectation from birth-order-dependent diversification within hemilineages.

(4) "We did not find any correlation between the morphology of premotor 13B and motor connections" - this needs to be explained, as morphology constrains connectivity.

We agree that morphology often constrains connectivity. However, in contrast to 13A neurons—where morphological clusters strongly predict MN connectivity—we did not observe such a correlation for 13B neurons. As we noted in our response to comment 2d, 13B neurons can form synapses onto MNs without exhibiting extensive or spatially structured overlap of their axonal projections with MN dendrites. This suggests that 13B → MN connectivity may be governed by more local, synapse-specific rules rather than by large-scale morphological positioning, in contrast to the spatially organized premotor map we observe for 13As.

(5) *"Based on their connectivity, we hypothesized that continuously activating them might reduce extension and increase flexion. Conversely, silencing them might increase extension and reduce flexion."* - these clear predictions are then not directly addressed in the results that follow.

We have now expanded this section.

(6) *"Thus, 13A neurons regulate both spatial and temporal aspects of leg coordination"*
"Together, 13A and 13B neurons contribute to both spatial and temporal coordination during grooming" - are these not intrinsically linked? This needs to be explained/justified.

The spatial (leg positioning, joint angles) and temporal (frequency, rhythm) aspects are often linked, but they can be at least partially dissociated. This has been shown in other systems: for example, Argentine ants reduce walking speed on uneven terrain primarily by decreasing stride frequency while maintaining stride length (Clifton et al., 2020), and *Drosophila* larvae adjust crawling speed mainly by modulating cycle period rather than the amplitude of segmental contractions (Heckscher et al., 2012). Consistent with these findings, we observe that 13A neuron manipulation in dusted flies significantly alters leg positioning without changing the frequency of walking cycles. Thus, leg positioning can be perturbed while the number of extension–flexion cycles per second remains constant, supporting the view that spatial and temporal features are at least partially dissociable.

(7) *"Connectome data revealed that 13B neurons disinhibit motor pools (...) One of these 13B neurons is premotor, inhibiting both proximal and tibia extensor MN"* - these are not possible at the same time.

We show that the 13B population contains neurons with distinct connectivity motifs:

some inhibit premotor 13A neurons (leading to disinhibition of motor pools), while others directly inhibit motor neurons. The split-GAL4 line we use labels three 13B neurons—two that inhibit the primary 13A neuron 13A-9d-γ (which targets proximal extensor and medial flexor MNs) and one that is premotor, directly inhibiting both proximal and tibia extensor MNs. Although these functions may appear mutually exclusive, their combined action could converge to a similar outcome: disinhibition of proximal extensor and medial flexor MNs while simultaneously inhibiting medial extensor MNs. This suggests that the labeled 13B neurons act in concert to bias the network toward a specific motor state rather than producing contradictory effects.

(8) *"we often observed that one leg became locked in flexion while the other leg remained extended, (indicating contribution from additional unmapped left right coordination circuits)." - Are these results not informative? I'd suggest the authors explain the implications of this more, rather than mentioning it within brackets like this.*

We agree with the reviewer that these results are highly informative. The observation that one leg can remain locked in flexion while the other stays extended suggests that additional left–right coordination circuits are engaged during grooming. This cross-talk is likely

mediated by commissural interneurons downstream of inhibitory premotor neurons, which have not yet been systematically studied. Dissecting these circuits will require a dedicated project combining bilateral connectomic reconstruction, studying downstream targets of these commissural neurons, and functional interrogation, which is beyond the scope of the current study.

(9) *"Indeed, we observe that optogenetic activation of specific 13A and 13B neurons triggers grooming movements. We also discover that" - this phrasing suggests that this has already been shown.*[external](#)

We replaced 'indeed' with "Consistent with this connectivity,"

(10) *"But the 13A circuitry can still produce rhythmic behavior even without those sensory inputs (or when set to a constant value), although the legs become less coordinated." - what does this mean?*

We can train (fine-tune) the model without the descending inputs from the "black box" and the behavior will still be rhythmic, meaning that our modeled 13A circuit alone can produce rhythmic behavior, i.e. the rhythm is not generated externally (by the "black box"). We added Figure 7 to the MS and re-wrote this paragraph. In the revised manuscript we now state: "But the 13A circuitry can still produce rhythmic behavior even without those excitatory inputs from the "black box" (or when set to a constant value), although the legs become less coordinated (because they are "unaware" of each other's position at any time). Indeed, when we refine the model (with the evolutionary training) without the "black box" (using instead a constant input of 0.1) the behavior is still rhythmic although somewhat less sustained ([Figure 7](#)). This confirms that the rhythmic activity and behavior can emerge from the modeled pre-motor circuitry itself, without a rhythmic input."

(11) *"However, to explore the possibility of de novo emergent periodic behavior (without the direct periodic descending input) we instead varied the model's parameters around their empirically obtained values." - why do the authors not show how the model performs without tuning it first? What are the changes exactly that are happening as a result of the tuning? Are there specific connections that are lost? Do I interpret Figure 6B and C correctly when I think that some connections are lost (e.g., an SN-MN connection)? How does that compare to the text, which states that "their magnitudes must be at least 80% of the empirical weights"?*

Without the fine-tuning we do not get any behavior (the activation levels saturate). So, we tolerate 20% divergence from the empirically established weights and we keep the signs the same. However, in the previous version we allowed the weights to decrease below 20% of the empirical weight (as long as the sign didn't change) but not above (the signs were maintained and synapses were not added or removed). We thank the reviewer for observing this important discrepancy. In the current version we ensured that the model's weights are bounded in both directions (the tolerance = 0.2), but we also partially relaxed the constraint on adjacency matrix re-scaling (see Methods, the "The fine-tuning of the synaptic weights" section, where we now clarify more precisely how the evolving model is fitted to the connectome constraints). We then re-ran the fine-tuning process. The Figure 6B and C is now corrected with the properly constrained model, as well as other panels in the figure. We also applied a better color scheme (now, blue is inhibitory and red is excitatory) for Fig. 6B and C.

(12) *"Interestingly, removing 13As-ii-MN connections to the three MNs (second row of the 13A → MN matrices in Figures 6B and C) does not have much effect on the leg movement (data not shown). It seems sufficient for this model to contract only one of the two*

antagonistic muscles per joint, while keeping the other at a steady state." - this is not clear.

We repeated this test with the newly fine-tuned model and re-wrote the result as follows: “...when we remove just the 13A-i-MN connections (which control the flexors of the right leg) we likewise get a complete paralysis of the leg. However, removing the 13A-ii-MN (which control the extensors of the right leg) has only a modest effect on the leg movement. So, we need the 13A-i neurons to inhibit the flexors (via motor neurons), but not extensors, in order to obtain rhythmic movements.”

(13) The Discussion needs to reference the specific Results in all relevant sections.

We have revised the discussion to explicitly reference the specific results.

(14) "Flexors and extensors should alternate" - there are circumstances in which flexors and extensors should co-contract. For instance, co-contraction modulates joint stiffness for postural stability and helps generate forces required for fast movements.

Thanks for pointing this out. We added “However, flexor–extensor co-contraction can also be functionally relevant, such as for modulating joint stiffness during postural stabilization or for generating large forces required for fast movements (Zakotnik et al., 2006; Günzel et al., 2022; Ogawa and Yamawaki 2025). Some generalist 13A neurons could facilitate co-contraction across different leg segments, but none target antagonistic motor neurons controlling the same joint. Therefore, co-contraction within a single joint would require the simultaneous activation of multiple 13A neurons.”

(15) "While legs alternate between extension and flexion, they remain elevated during grooming. To maintain this posture, some MNs must be continuously activated while their antagonists are inactivated." - this is not necessarily correct. Small limbs, like those of Drosophila, can assume gravity-independent rest angles (10.1523/JNEUROSCI.5510-08.2009).

We added it to discussion

(16) The discussion "Spatial Mapping of premotor neurons in the nerve cord" seems to me to be making obvious points, and does not need to be included.

We have now revised this section to highlight the significance of 13A spatial organization, emphasizing premotor topographic mapping, multi-joint movement modules, and parallels to myotopic, proprioceptive, and vertebrate spinal maps.

(17) Key point, albeit a small one: "Normal activity of these inhibitory neurons is critical for grooming" - the use of the word critical is problematic, and perhaps typical of the tone of the manuscript. These animals still groom when many of these neurons are manipulated, so what does "critical" really mean?

In this instance, we now changed “critical” to “important”. We observed that silencing or activating a large number (>8) 13A neurons or few 13A and B neurons together completely abolishes grooming in dusted flies as flies get paralyzed or the limbs get locked in extreme poses. Therefore we think we have a justification for the statement that these neurons are critical for grooming. These neurons may contribute to additional behaviors, and there may be partially redundant circuits that can also support grooming. We have revised the manuscript with the intention of clarifying both what we have observed and the limits.

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